Supplementary Material for "The Wrong Unit of Uncertainty: Simultaneous Inference for Repeated Cross-National Surveys"

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This online supplement collects secondary material referenced from the main manuscript. All content, numbers, and figures are reproduced unchanged; only figure numbers carry an "S" prefix.

S1 Proofs

This section gives full statements and proofs of the results of the main text, in the main-manuscript numbering. Each result is paired with an automated contract test in the replication package that checks its computational implications numerically. Where the source argument is a sketch or rests on an estimated constant, we say so.

Common setup. We observe K exchangeable source countries $c = 1, \ldots, K$ plus a target $K{+}1$. Each country c has a latent response-distribution curve F_c (a CDF over a grid $t_1 <$

$\cdots < t_T$, stacked over rounds where a trajectory is meant) and a transport center μ (a leave-one-out mean or fitted center). Three curves are kept apart, in the main text's symbols: the latent deviation $G_c = F_c - \mu$; the design error S_c of the complex-survey estimate, $\widetilde{F}_c = F_c + S_c$, with coordinate-wise design variance $v_c(t)^2$ estimated by a stratified-PSU / replicate-weight bootstrap; and the observed deviation $\widetilde{G}_c = \widetilde{F}_c - \mu = G_c + S_c$. The latent transport score is $R_c = \max_t |G_c(t)|/\sigma(t)$ and the observed score $\widetilde{R}_c = \max_t |\widetilde{G}_c(t)|/\sigma(t)$ ($\sigma \equiv 1$ for the deployed unstudentized anchors); $\xi_c := \widetilde{R}_c - R_c$ is the score contamination, a difference of maxima with $|\xi_c| \le \max_t |S_c(t)|/\sigma(t)$, on which no centring or variance structure is assumed anywhere below. Design noise is quantified at the coordinate, where the additive model is exact: $s^2_{\text{plug}}(t) = \operatorname{Var}_c \widetilde{G}_c(t) = s^2_R(t) + \overline{v^2}(t)$ for design error independent of the latent curve, and the design share $\rho(t)^2 = \overline{v^2}(t)/s^2_{\text{plug}}(t) \in [0, 1)$; the scalar $\widehat{\rho}_{\text{LCB}}$ of the selector is the conservative aggregate of this profile defined in `pcb.inference.design_aware.rho_lcb` (ratio of t-averaged bounds). Two targets are kept distinct throughout: the survey-estimate score $\widetilde{R}_{K+1}$ (T1) and the latent score R_{K+1} (T2), the object the political claim concerns.

S1.1 Theorem 1 (non-identification and necessity of design information)

Statement (curve level). The result is stated for the curve-level model $\widetilde{G}_c = G_c + S_c$ of the main text's decomposition (1): the latent deviation curves $G_1, \dots, G_K, G_{\text{new}} \in \mathbb{R}^T$ are exchangeable with law $\mathcal{L}_G$; the design-noise curves S_c are i.i.d. with law $\mathcal{S}$, independent of the latent curves (the two-level survey structure: given the drawn population, the sampling mechanism is independent of which curve was drawn); and $\mathcal{S}$ is known only to lie in an admissible class $\mathfrak{S}$ that (a) contains δ_0 (negligible design noise) and (b) is closed under one further convolution by some nondegenerate symmetric law $\mathcal{N}$, with $\mathcal{L}_G$ likewise expressible with and without the factor $\mathcal{N}$. No additive or independence structure is assumed on the sup-scores, and $\mathcal{S}$ may be arbitrarily heteroskedastic across thresholds (e.g. the $F(t)(1 - F(t))$ sampling variance). Let $\widehat{q}_{\text{plug}}$ be the $\lceil (1 - \alpha)(K+1) \rceil$-th order statistic of $\{\|\widetilde{G}_c\|_\infty\}$, the plug-in radius of Theorem 3 built on the observed curves, with $\widehat{q}_{\text{plug}} = \infty$ when that rank exceeds K (Theorem 3's convention; the statement is then trivial). (i) $\mathcal{L}_G$ is not identified from $\mathcal{L}_{\widetilde{G}} = \mathcal{L}_G * \mathcal{S}$ over $\mathfrak{S}$. (ii) Write $s_K := \lceil (1-\alpha)(K{+}1) \rceil/(K{+}1) - (1-\alpha) \geq 0$ for the conformal discreteness slack. Let $\widehat{\rho}$ be a non-randomized measurable functional of the observed curves alone with $\widehat{\rho} \leq \widehat{q}_{\text{plug}}$ almost surely, and suppose the band $\mu \pm \widehat{\rho}$ is valid at level $1-\alpha$ for the latent target under *every* $\mathcal{S} \in \mathfrak{S}$. Then on the member $\mathcal{S} = \delta_0$ the coverage deficit relative to the plug-in is bounded by the slack, $\Pr\left(\widehat{\rho} < \|G_{\text{new}}\|_\infty \leq \widehat{q}_{\text{plug}}\right) \leq s_K$: no observed-curve band valid over the class improves on the plug-in uniformly by more than the discreteness slack. A randomized band can shave exactly s_K and no more. (The statement bounds uniform improvement; a band narrower on some realizations and wider on others is not excluded, which is why we phrase the main-text reading as "no observed-curve band can beat the plug-in *uniformly*, up to the discreteness slack." By Remark S1 no observed-curve band—including the plug-in—is latent-valid over *all* admissible laws, so validity must be read relative to the stated subclass.)

Proof. (i) Convolution in $\mathbb{R}^T$ is not injective on the admissible class. With $\mathcal{N}$ as in condition (b), pick $\mathcal{S}_0 \in \mathfrak{S}$ with $\mathcal{S}_0 * \mathcal{N} \in \mathfrak{S}$ and put $\mathcal{L}'_G = \mathcal{L}_G * \mathcal{N}$, $\mathcal{S} = \mathcal{S}_0 * \mathcal{N}$, $\mathcal{S}' = \mathcal{S}_0$. Both pairs are admissible, distinct ($\mathcal{N}$ nondegenerate), and $\mathcal{L}_G * \mathcal{S} = \mathcal{L}_G * \mathcal{S}_0 * \mathcal{N} = \mathcal{L}'_G * \mathcal{S}'$: the same observed law, with the spread split differently between latent signal and design noise. (A single coordinate $T{=}1$ already suffices, e.g. Gaussian $N(0,\sigma^2-\tau^2) * N(0,\tau^2) = N(0,\sigma^2)$ for any $0 < \tau^2 < \sigma^2$; the general-T statement follows by product laws.)

(ii) A radius $\widehat{\rho} = \widehat{\rho}(\{\widetilde{G}_c\})$ has the same joint law with the observed data in any two configurations sharing the observed law—in particular the two of part (i), whose latent laws differ by the dispersing factor $\mathcal{N}$. Validity for every admissible $\mathcal{S}$ therefore forces the radius to cover the most-dispersed admissible latent law compatible with the observed data, attained at $\mathcal{S} = \delta_0$, where all observed spread is latent. Concretely, take $\mathcal{S} = \delta_0 \in \mathfrak{S}$. Then $\widetilde{G}_c = G_c$ for every c, and the target satisfies $F_{\text{new}} \in \mu \pm \widehat{\rho} \iff \|G_{\text{new}}\|_\infty \leq \widehat{\rho}$, with $\|G_{\text{new}}\|_\infty$ exchangeable with the K observed sup-scores. The conformal rank identity gives $\Pr\left(\|G_{\text{new}}\|_\infty \le \widehat{q}_{\text{plug}}\right) = \lceil(1-\alpha)(K+1)\rceil/(K+1) = (1-\alpha) + s_K$, so a non-randomized radius sitting almost surely at or below $\widehat{q}_{\text{plug}}$ has latent coverage $(1-\alpha) + s_K - \Pr\big(\widehat{\rho} < \|G_{\text{new}}\|_\infty \le \widehat{q}_{\text{plug}}\big)$; validity at $1-\alpha$ on this member therefore forces the deficit to be at most s_K, which is the claimed bound. The slack is attainable and no more: randomized smoothing of the conformal rank (randomizing between adjacent order statistics) achieves latent coverage exactly $1-\alpha$ on this member, and any band shaving more than s_K of coverage drops below $1-\alpha$ there. Strict efficiency beyond the slack is therefore attainable only by shrinking $\mathfrak{S}$—i.e. by supplying information on the law of S (equivalently the v_c), which is exactly the design-bootstrap information the deconvolution branch estimates (Theorem 4), and whose finite-K reliability is what Proposition 2 bounds. □

S1.2 Theorem 2 (oracle validity)

Statement. With the countries exchangeable and the design law known: (a) the clustered band on $\{\widetilde{R}_c\}$ covers the survey-estimate target T1 exactly, $\Pr(\widetilde{R}_{K+1} \le \widehat{q}) \ge 1-\alpha$, using exchangeability only. (b) For the latent target T2 the same band satisfies, for every $u > 0$,

$$\Pr(R_{K+1} \le \widehat{q}) \;\ge\; 1-\alpha \;-\; \Pr(\xi_{K+1} < -u) \;-\; \Pr\left(\widetilde{R}_{K+1} \in (\widehat{q}-u,\ \widehat{q}]\right),$$

so the T1→T2 coverage gap is controlled by a noise-tail term and a local anti-concentration term, both of which vanish with the design-noise scale: since $|\xi_{K+1}| \le \|S_{K+1}\|_\infty$, the tail term is at most $\Pr(\|S_{K+1}\|_\infty > u)$, a statement about the design error alone, whatever the shape of the score law. Quantitatively, write σ_ξ for a scale of ξ (a bound on its standard deviation, itself at most the sup-norm scale of S). With countries independent (so that the target score is independent of $\widehat{q}$) and a density bound L for $\widetilde{R}$ near its upper quantile, optimizing u under a variance-only tail bound (Chebyshev) gives a gap of order $\sigma_\xi^{2/3}$; under sub-Gaussian design-noise tails—the natural case for bootstrap-estimated design error—the optimized gap is $O(\sigma_\xi\sqrt{\log(1/\sigma_\xi)})$. We use independence, not mere exchangeability, for the anti-concentration step; the tail term needs neither, and neither term assumes ξ centred. Equivalently, the *noise-inflated* band with radius $\widehat{q} + u$ covers the latent target at level $1 - \alpha - \Pr(|\xi_{K+1}| > u)$ with no anti-concentration condition. (c) Under the scale-family assumption (A1) of Theorem 4, the oracle deconvolution band with the true target scale is finite-sample valid for the latent target, exact at the attainable conformal rank (Theorem 4(i)), and its target scale is, coordinate by coordinate, a factor $\sqrt{1 - \rho(t)^2}$ of the observed scale $s_{\text{plug}}(t)$; the realized band width carries in addition the ratio of the two conformal quantiles, which tends to one as $\rho \to 0$ (AW-1) but is not a function of $\rho(t)$. Both statements live under (A1): the variance identity $s_T^2(t) = s_{\text{plug}}^2(t)(1-\rho(t)^2)$ holds without it, but turning a scale identity into a band-width ratio and a latent-coverage statement requires the standardized shapes to match, which is what (A1) supplies and what Remark S1 shows cannot be dispensed with.

Proof. (a) By exchangeability of $\{\widetilde{R}_c\}_{c=1}^{K+1}$ the rank of $\widetilde{R}_{K+1}$ is uniform on $\{1, \ldots, K+1\}$; with $\widehat{q}$ the $\lceil (1-\alpha)(K+1) \rceil$-th order statistic, $\Pr(\widetilde{R}_{K+1} \leq \widehat{q}) = \lceil (1-\alpha)(K+1) \rceil / (K+1) \geq 1 - \alpha$, finite-sample and distribution-free, using no assumption on the design-noise law. (b) On $\{R_{K+1} > \widehat{q}\}$, either $\xi_{K+1} < -u$ or $\widetilde{R}_{K+1} = R_{K+1} + \xi_{K+1} > \widehat{q} - u$. Hence $\Pr(R_{K+1} > \widehat{q}) \leq \Pr(\xi_{K+1} < -u) + \Pr(\widetilde{R}_{K+1} > \widehat{q} - u)$, and $\Pr(\widetilde{R}_{K+1} > \widehat{q} - u) \leq \alpha + \Pr(\widetilde{R}_{K+1} \in (\widehat{q} - u, \widehat{q}])$ by part (a). For the inflated band, $\{R_{K+1} > \widehat{q} + u\} \subseteq \{\widetilde{R}_{K+1} > \widehat{q}\} \cup \{|\xi_{K+1}| > u\}$ directly. (c) At each coordinate the design error is mean-zero (design unbiasedness of the survey estimate) and independent of the latent curve, so $\operatorname{Var}_c \widetilde{G}_c(t) = \operatorname{Var}_c G_c(t) + \overline{v^2}(t)$ exactly, i.e. $s_T^2(t) = s_{\text{plug}}^2(t)\,(1 - \rho(t)^2)$ for the oracle deconvolution scale, so the scale factor $s_R(t)/s_{\text{plug}}(t) = \sqrt{1 - \rho(t)^2}$ is an identity. Under (A1) the deconvolution band's coordinate radius is $q_D\, s_R(t)$, against $q_S\, s_{\text{plug}}(t)$ for a studentized anchor and the constant q_P for the deployed unstudentized anchor; the realized width ratios, $(q_D/q_S)\sqrt{1 - \rho(t)^2}$ and $(q_D/q_P)\, s_R(t)$, therefore carry a quantile ratio that the scale identity does not determine (AW-1 shows only that it tends to one as $\rho \to 0$). No statement about the variance of the sup-score $\widetilde{R}$ is used. $\square$

Remark S1 (Symmetric noise alone is *not* sufficient for latent conservativeness). It is tempting to argue that symmetric design noise makes the observed-score band conservative for the latent target, on the grounds that a symmetric convolution is a mean-preserving spread. That inference is false without shape conditions on the latent score law: a mean-preserving spread does not imply quantile dominance at a fixed level. Counterexample: $R \in \{0, 10\}$ with masses $(0.85, 0.15)$ and $\xi = \pm 1$ with equal probability give $Q_{0.9}(R) = 10$ but $Q_{0.9}(R+\xi) = 9$ — the symmetric noise moves mass *below* the latent quantile — and the conformal band calibrated on the noisy scores undercovers the latent target (0.867 at $K = 200$, $\alpha = 0.10$, in the replication package's contract test `test_latent_target_bridge.py`). Bimodal latent score laws of this shape arise naturally when a minority of countries follows a distinct regime. This is why Theorem 2(b) is stated as an explicit two-term bound (small noise $\Rightarrow$ small gap) rather than as unconditional conservativeness, and why on real data we report the design-noise scale ($\widehat{\rho} \leq 0.22$ at the national level across all surveys examined, and ≤ 0.29 including the noisiest ESS age-band subgroups; Section S4) that makes the gap negligible rather than assuming it away. Under additional shape conditions (e.g. a unimodal latent law with $\widehat{q}$ above its mode, or a log-concave latent density) the classical dominance argument does go through, but we do not rely on it.

S1.3 Theorem 3 (exact finite-K validity of the deployed base band)

Statement. Let the error curves be $\widetilde{G}_c(t) = \widetilde{F}_c(t) - \mu(t)$, where the center μ satisfies the *symmetric-construction condition*: it is (i) a fixed function independent of the $K{+}1$ curves, or (ii) computed from a disjoint center fold, or (iii) computed *per unit* from that unit's own auxiliary data by a common rule (e.g. the country's own previous round, as in the deployed ESS certification), or (iv) a symmetric function of *all* $K{+}1$ curves including the target's (the surveyed-target validation mode). With the unstudentized cluster score $\widetilde{R}_c = \max_t |\widetilde{G}_c(t)|$ and the countries exchangeable, for *every* K the band $\widehat{B} = \text{center} \pm \widehat{q}$ with $\widehat{q}$ the $\lceil (1-\alpha)(K+1) \rceil$-th order statistic of $\{\widetilde{R}_c\}$ satisfies $\Pr(\widetilde{F}_{\text{new}} \in \widehat{B}) \geq \lceil (1-\alpha)(K+1) \rceil / (K+1) \geq$

$1-\alpha$, distribution-free—for the *survey-estimate* trajectory T1, since the scores are built from estimate-scale curves; the latent target T2 carries in addition the two-term gap of Theorem 2(b), or is reached directly under (A1) by the anchor-domination lemma below.

Proof. Under any of (i)–(iv) the map from the $K{+}1$ curves to the $K{+}1$ scores is symmetric in the units (conditional on the fold in case (ii)), so the scores $\widetilde{G}_1, \ldots, \widetilde{G}_{K+1}$ are exchangeable elements of $\mathbb{R}^T$. If the score law is atomless the $\{\widetilde{R}_c\}$ are a.s. distinct; with atoms (possible for discrete survey scales), break ties by a symmetric uniform randomization—the standard conformal device—which preserves exchangeability of the ranks, and the guarantee holds as stated (without randomization, ties can only make the band conservative, never anti-conservative, since the order statistic weakly increases). Therefore the rank of $\widetilde{R}_{K+1}$ is uniform on $\{1, \ldots, K{+}1\}$, and with $\widehat{q} = \widetilde{R}_{(m)}$, $m = \lceil (1-\alpha)(K{+}1) \rceil$, $\Pr(\widetilde{R}_{K+1} \leq \widehat{q}) = m/(K{+}1) \geq 1-\alpha$. The simultaneous coverage event is pointwise equivalent to the scalar event: $\widetilde{F}_{\text{new}}(t) \in \widehat{B}(t)\ \forall t \iff |\widetilde{G}_{K+1}(t)| \leq \widehat{q}\ \forall t \iff \max_t |\widetilde{G}_{K+1}(t)| \leq \widehat{q} \iff \widetilde{R}_{K+1} \leq \widehat{q}$. Hence simultaneous coverage is $\geq \lceil (1-\alpha)(K{+}1) \rceil/(K{+}1)$, distribution-free. Intersecting the band with $[0,1]$ and applying the isotonic tightening $L(t) = \text{clip}_{[0,1]}(\max_{t' \leq t} \text{lo}(t'))$, $U(t) = \text{clip}_{[0,1]}(\min_{t' \geq t} \text{hi}(t'))$ only shrink the band while still containing the monotone $[0,1]$-valued truth, preserving validity. The split-modulation studentized variant (modulation computed on a disjoint fold) is exact by the identical argument conditional on the modulation fold. □

Remark S2 (Transport centers estimated from the calibration pool). A grand mean over the calibration countries, used as the center for an *unsurveyed* target, violates the symmetric-construction condition: each calibration country sits inside its own center (deviations shrunk by $(K{-}1)/K$) while the target sits outside, which biases $\widehat{q}$ downward — an anti-conservative $O(1/K)$ asymmetry. The deployed remedy is the leave-one-out deviation $\widetilde{G}_c = \widetilde{F}_c - \text{mean}_{c' \neq c} \widehat{\theta}_{c'}$ (`pcb.inference.conformal_band.loo_deviations`): each calibration score then excludes its own unit exactly as the target's score excludes the target. The residual asymmetry (a pool of $K{-}1$ versus K units) is covered by the following result, which places the deployed construction itself under a finite-sample guarantee.

Proposition S1 (Finite-sample validity of the leave-one-out deployment)**.** *Let $X_1, \dots, X_K$ be the calibration curves and X_{K+1} the target, exchangeable in $\mathbb{R}^T$; let $a_c = \|X_c - \bar{X}_{-c}\|$ be the deployed leave-one-out scores, with $\bar{X}_{-c}$ the mean of the other $K-1$ calibration curves and $\|\cdot\|$ any norm; let $R^* = \|X_{K+1} - \bar{X}\|$ be the target score, $\bar{X}$ the mean of the K calibration curves; and let $\widehat{q}$ be the $\lceil(1-\alpha)(K+1)\rceil$-th order statistic of $\{a_c\}$, with the convention $\widehat{q} = \infty$ when $\lceil(1-\alpha)(K+1)\rceil > K$ (as the implementation returns). Then, distribution-free and for every $K \geq 2$:* (i) $\Pr\big(R^* \leq \frac{K}{K-1}\widehat{q}\big) \geq \lceil(1-\alpha)(K+1)\rceil/(K+1) \geq 1-\alpha$ *— the deployed band inflated by $K/(K-1)$ is finite-sample valid;* (ii) *the uninflated deployed band satisfies* $\Pr(R^* \leq \widehat{q}) \geq 1 - \alpha - \delta_K$ *with* $\delta_K = \Pr\left(\widehat{q} < R^* \leq \frac{K}{K-1}\widehat{q}\right)$.

Proof. For $i = 1, \dots, K+1$ define the fully symmetric leave-one-out residuals over *all* $K+1$ units, $e_i = X_i - \mathrm{mean}_{j\neq i} X_j = \frac{K+1}{K}(X_i - \bar{X}_+)$, with $\bar{X}_+$ the mean of all $K+1$ curves. The array $(e_1, \dots, e_{K+1})$ is a symmetric function of the sample, so $(\|e_i\|)$ is exchangeable, and the conformal rank inequality gives $\Pr\left(\|e_{K+1}\| \leq \|e\|_{(m)}\right) \geq m/(K+1)$ for $m = \lceil(1-\alpha)(K+1)\rceil$, where $\|e\|_{(m)}$ is the m-th order statistic of the K calibration values. Two identities connect this infeasible construction to the deployed one. First, $X_{K+1} - \bar{X}_+ = \frac{K}{K+1}(X_{K+1} - \bar{X})$, so $\|e_{K+1}\| = R^*$ *exactly*: the target scores of the two constructions coincide. Second, for $c \leq K$, $e_c = \frac{K+1}{K}(X_c - \bar{X}) - \frac{1}{K}(X_{K+1} - \bar{X})$, while $a_c = \frac{K}{K-1}\|X_c - \bar{X}\|$; since $\frac{K+1}{K} \leq \frac{K}{K-1}$, the triangle inequality gives $\|e_c\| \leq a_c + R^*/K$ for every c. On the event $\{R^* \leq \|e\|_{(m)}\}$ at least $K - m + 1$ calibration units have $\|e_c\| \geq R^*$, hence $a_c \geq R^*(1 - 1/K)$ for those units, hence $\widehat{q} \geq R^* \frac{K-1}{K}$, which is (i); when $\lceil(1-\alpha)(K+1)\rceil > K$ the convention makes both radii infinite and (i) is trivial. Part (ii) is immediate from (i), the window $(\widehat{q}, \frac{K}{K-1}\widehat{q}]$ having relative width $1/(K-1)$. □

Remark S3 (Rate for the uninflated remainder; the default is not to need one). The window in δ_K has $\mathcal{F}_K$-measurable endpoints, where $\mathcal{F}_K$ is the calibration σ-field generated by $X_1, \dots, X_K$ (both $\widehat{q}$ and the centering $\bar{X}$ inside R^* are functions of it), so a bound on the *marginal* density of R^* does not control it. The correct assumption is conditional: if the conditional law of R^* given $\mathcal{F}_K$ has density at most L a.s. on a neighbourhood of the deployed radius, and $\widehat{q} \le \bar{q}$ a.s., then $\Pr\left(\widehat{q} < R^* \le \frac{K}{K-1}\widehat{q} \mid \mathcal{F}_K\right) \le L\,\widehat{q}/(K-1)$ a.s., and taking expectations, $\delta_K \le L\,\bar{q}/(K-1) = O(1/K)$. The released implementation does not lean on this rate: its default deployment ships the inflated radius of part (i), which needs no anti-concentration assumption at all, and the uninflated remainder survives only as an efficiency remark.

The remainder δ_K is measured directly in the replication package (`e58`): across four families (Gaussian, t_3, skewed, dependent-round) and $K \in \{6, \ldots, 60\}$, 28 cells at 20,000 replications each, uninflated leave-one-out coverage never falls below the exact-construction floor by more than twice the Monte Carlo standard error — the measured δ_K is zero — while the grand-mean construction the remark above excludes shows its real $O(1/K)$ deficit (worst -0.064), and a split-fold center returns an infinite radius whenever halving drops the calibration fold below $\lceil 1/\alpha \rceil$ units, i.e. throughout the cross-national range $K \le 15$ at $\alpha = 0.10$. The asymmetry vanishes entirely under constructions (i)–(iv), which the headline ESS certification satisfies through its own-previous-round centers (case iii). The $K/(K-1)$-inflated deployment of Proposition S1(i) is the released *default* (`dapcb` with `loo_center=True`; factor `loo_exact_inflation`): the frozen gates are evaluated on the uninflated radii, exactly the decision rule the sealed validation runs froze, and the inflation is a monotone widening applied to the returned anchor radius only — so every sealed coverage result is inherited a fortiori, and the cost is $1/(K-1)$ of width: 3.5% at the ESS's $K \approx 30$, 5% at the AmericasBarometer's $K \approx 20$.

Modulation self-inclusion deficit (proof sketch; estimated constant). If instead one studentizes by an in-sample modulation $\widehat{s}(t) = \mathrm{SD}_c\,\widetilde{G}_c(t)$ computed from the calibration curves themselves, the unobserved target does not share this denominator, which breaks exchangeability. A first-order expansion of the order-statistic bias in the number g of modulation slots each country contributes to gives a coverage deficit $\approx 0.382\,(g/K)$; the constant 0.382 is estimated (empirical $R^2 = 0.923$), not derived, and is flagged as such. Pooled modulation ($g = 1$) is safe; slotwise ($g = L$) undercovers by $\approx 0.382\,L/K$. This $O(g/K)$ deficit is why the deployed band is unstudentized.

Lemma S1 (Scalar-modulation invariance)**.** *If the modulation is a single in-sample* scalar $\widehat{s}$ *(the same number at every threshold and slot), the studentized band is* identical *to the unstudentized band, hence exact at any K: the scores are $\max_t |\widetilde{G}_c(t)|/\widehat{s}$, their m-th order statistic is $M_{(m)}/\widehat{s}$ with $M_{(m)}$ the unstudentized order statistic, and the band radius $\widehat{q} \cdot \widehat{s} = M_{(m)}$ — the data-dependent scalar cancels identically. Self-inclusion deficits therefore arise* only *from non-constant modulation profiles $\widehat{s}(t)$, whose* shape *(not level) is what the unobserved target fails to share. This sharpens the note above: the $O(g/K)$ penalty is the price of profile estimation, and the deployed U0 band (U0: the unstudentized clustered construction, the paper's anchor branch) gives up only the profile adaptivity — the level normalization was free all along.*

S1.4 Theorem 4 (estimated-law validity, with explicit assumptions)

This section states the main text's Theorem 4 with its three assumptions in full and proves it. Under them the oracle case is *exact* and the estimated case carries a proved, one-sided remainder. The assumptions are not decorative — the bimodal counterexample of Remark S1 shows that *some* shape restriction is necessary for any latent-target claim, and (A1) is exactly the restriction under which the deconvolution's distributional matching is an identity rather than an approximation.

Assumptions. (A1, scale family.) There exist i.i.d. random curves $W_1, \ldots, W_{K+1}$ and deterministic-given-design scales such that the observed calibration curves satisfy $\widetilde{G}_c(t) = \sqrt{s_R(t)^2 + v_c(t)^2}\, W_c(t)$ and the latent target curve satisfies $G_{K+1}(t) = s_R(t)\, W_{K+1}(t)$, with W normalized coordinate-wise to $\mathbb{E}\, W(t) = 0$ and $\mathbb{E}\, W(t)^2 = 1$ (so that s_R^2 and v_c^2 are variances and $s_{\text{plug}}^2 = s_R^2 + \overline{v^2}$ is the main text's coordinate identity) and the latent scale nondegenerate and bounded, $0 < s_{\min} \le s_R(t) \le s_{\max} < \infty$ for every t (the relative perturbations $\Delta_0(t)$ below divide by $s_R(t)$). This is strong, and we state its content plainly: it requires not only a common (e.g. Gaussian) shape family but that the design noise share the latent curves' *cross-*

threshold correlation profile, so that noise changes only the scale of a common shape process. In real surveys the within-country sampling covariance across thresholds is governed by the design while the between-country profile is governed by substantive heterogeneity, so (A1) is an idealization—the price of finite-sample exactness. Bimodal mixtures of Remark S1 violate it. *(A2, anti-concentration.)* Two i.i.d. sup-scores of the shape process enter the proofs: $M := \max_t |W(t)|$ (deconvolution branch, at its $(1-\alpha_{\text{dec}})$ quantile q^*) and the latent anchor score $R^{\text{lat}} := \max_t s_R(t)|W(t)|$ (anchor branches, at its $(1-\alpha_{\text{anchor}})$ quantile q^{lat}). Each has a density bounded by L on a fixed h_0-neighbourhood of its quantile and puts mass at least $\eta > 0$ on each side of it inside that neighbourhood, $F(q^-) - F(q - h_0) \geq \eta$ and $F(q + h_0) - F(q) \geq \eta$ — the condition under which its order statistic concentrates there (Lemma S2). A continuous score with positive density at the quantile satisfies both for small h_0. Nothing is assumed about the heterogeneous observed scores $\widetilde{R}_c = \max_t \sqrt{s_R^2 + v_c^2}\,|W_c|$, which are independent but not identically distributed across countries; the proofs reach them only through pathwise domination. *(A3, moments, boundedness, and bootstrap.)* The scores have finite fourth moment (kurtosis κ) and the error curves are uniformly bounded (automatic on the CDF scale, where $|\widetilde{G}_c(t)| \leq 1$), so that empirical second moments and bootstrap replicate averages obey Bernstein-type sub-exponential tail bounds; and the size-B design bootstrap is unbiased for $v_c^2(t)$ with relative variance $\leq c_v/B$. (Boundedness is what buys the logarithmic—rather than polynomial—dependence on KT below; without it, Chebyshev alone gives the same conclusion with $\log(KT)$ replaced by a polynomial factor and a correspondingly slower rate.)

Lemma S2 (Order-statistic concentration)**.** *Let $Z_1, \ldots, Z_K$ be i.i.d. with distribution function F, let $q = \inf\{x : F(x) \geq 1-\alpha\}$, $m = \lceil (1-\alpha)(K+1) \rceil$, and suppose $F(q^-) - F(q-h) \geq \eta$ and $F(q+h) - F(q) \geq \eta$ for some $h, \eta > 0$. Then for $K > 2/\eta$,*

$$\Pr\left(|Z_{(m)} - q| > h\right) \;\leq\; 2\exp\big\{-2K(\eta - 2/K)^2\big\},$$

so $r_K := \Pr(|Z_{(m)} - q| > h) = O(e^{-cK})$*, in particular* $o(K^{-1/2})$.

Proof. $Z_{(m)} > q+h$ iff fewer than m of the Z_i are $\le q+h$, i.e. $\widehat{F}_K(q+h) < m/K$. Since $m \le (1-\alpha)(K{+}1)+1$, $m/K \le 1-\alpha+2/K$, while $F(q+h) \ge F(q)+\eta \ge 1-\alpha+\eta$; hence $F(q+h) - \widehat{F}_K(q+h) > \eta - 2/K$, an event of probability at most $\exp\{-2K(\eta-2/K)^2\}$ by the one-sided Dvoretzky–Kiefer–Wolfowitz–Massart inequality. Symmetrically, $Z_{(m)} < q-h$ iff at least m of the Z_i are $< q-h$, i.e. $\widehat{F}_K((q-h)^-) \ge m/K \ge 1-\alpha$, while $F(q-h) \le F(q^-) - \eta \le 1-\alpha-\eta$; the deviation exceeds η, with probability at most $\exp\{-2K\eta^2\}$. Add the two. $\square$

Statement. Under (A1)–(A3), the always-deconvolve band — radius $\widehat{q}\cdot\widehat{s}_T(t)$ with $\widehat{q}$ the $\lceil(1-\alpha_{\text{dec}})(K{+}1)\rceil$-th order statistic of the studentized scores $\max_t |\widetilde{G}_c(t)|/\sqrt{\widehat{s}_T(t)^2 + \widehat{v}_c(t)^2}$ — satisfies: (i) with *oracle* scales ($\widehat{s}_T = s_R$, $\widehat{v}_c = v_c$) it is finite-sample valid for the latent target at every K, and *exact* at the attainable conformal rank whenever $\lceil(1{-}\alpha)(K{+}1)\rceil \le K$ and the scores are a.s. distinct; (ii) with estimated scales,

$$\Pr\left(F_{\text{new}} \in \widehat{B}\right) \;\ge\; 1{-}\alpha_{\text{dec}}{-}\underbrace{2Lq^*\,\delta_{K,B}}_{\text{scale error}} - \underbrace{\eta_{K,B}}_{\text{tail event}} - \underbrace{r_K}_{\text{order-stat}}, \qquad \delta_{K,B} = C_1\sqrt{\tfrac{(\kappa-1+z^2)\log K}{K}} + C_2\sqrt{\tfrac{c_v \log(KT)}{B}},$$

with $\eta_{K,B} = O(K^{-1/2})$ obtainable from the Bernstein bounds of (A3), $r_K = O(e^{-cK})$ from Lemma S2, and order-statistic concentration (C_1, C_2 depend on (κ, z, L, q^*) through those inequalities; we do not display numerical values), so $\varepsilon_{K,B} := 2Lq^*\delta_{K,B} + \eta_{K,B} + r_K \to 0$ at rate $\sqrt{\log(KT)/\min(K,B)}$; and (iii) the finite-$K$ guard makes the design-variance-subtraction component of the scale error one-sided: on the guard event $\mathcal{E}_{\text{guard}} = \{[\,\overline{\widehat{v}^2}(t) - z\ \mathrm{SE}(t)\,]_+ \le \overline{v^2}(t)\ \forall t\}$ the guard subtracts *less* design variance than the truth, so $\widehat{s}^2_T(t) \ge \max\{\widehat{s}^2_{\text{plug}}(t) - \overline{v^2}(t), \text{floor}\}$ and that component errs wide, not narrow. No theorem-level probability is attached to $\mathcal{E}_{\text{guard}}$ for the deployed normal-reference $z = 1.645$; the remaining, two-sided sampling error in $\widehat{s}^2_{\text{plug}}$ is what (ii)'s $\varepsilon_{K,B}$ controls.

Proof. (i) Under (A1) with oracle scales the studentized calibration scores are $\max_t |\widetilde{G}_c|/\sqrt{s_R^2 + v_c^2} =$

$\max_t |W_c| = M_c$, i.i.d., and the target's studentized score is $\max_t |G_{K+1}|/s_R = M_{K+1}$, an independent copy. The $K+1$ scores are exchangeable, the rank of M_{K+1} is uniform, and the order-statistic band is exact — the argument of Theorem 3 verbatim. (This also proves the oracle clause of Theorem 2(c).)

(ii) Define the multiplicative perturbations $\Delta_c(t) = \sqrt{\widehat{s}_T^2 + \widehat{v}_c^2} \big/ \sqrt{s_R^2 + v_c^2} - 1$ and $\Delta_0(t) = \widehat{s}_T(t)/s_R(t) - 1$, and $\delta^* = \max\big(\max_{c,t} |\Delta_c(t)|, \max_t |\Delta_0(t)|\big)$. On $\{\delta^* \le \delta\}$ every studentized calibration score is sandwiched, $M_c/(1+\delta) \le \widehat{S}_c \le M_c/(1-\delta)$ (a uniform multiplicative bound on the denominator survives the $\max_t$), hence $\widehat{q} \ge M_{(m)}/(1+\delta)$; and the target's statistic $\max_t s_R W_{K+1}/\widehat{s}_T \le M_{K+1}/(1-\delta)$. A miss therefore implies $M_{K+1} > M_{(m)}(1-\delta)/(1+\delta) \ge M_{(m)}(1-2\delta)$, so

$$\Pr(\text{miss}) \;\le\; \underbrace{\Pr\big(M_{K+1} > M_{(m)}\big)}_{\le \alpha_{\text{dec}} \text{ (exchangeable ranks)}} \;+\; \Pr\big(M_{K+1} \in (M_{(m)}(1-2\delta),\, M_{(m)}]\big) \;+\; \Pr(\delta^* > \delta).$$

For the middle term, $M_{(m)}$ concentrates on q^*: by Lemma S2 under (A2)'s mass condition, $r_K := \Pr(|M_{(m)} - q^*| > h_0/2) = O(e^{-cK})$. The two radii are different objects and are kept apart: h_0 is (A2)'s fixed neighbourhood, δ the shrinking scale-estimation error (set to $\delta_{K,B}$ below). On the complementary event, and for K, B large enough that $2q^*\delta_{K,B} \le h_0/2$, the perturbation window $(M_{(m)}(1-2\delta), M_{(m)}]$, of length $\le 2\delta q^*(1+o(1))$, lies inside the h_0-neighbourhood of q^* where (A2)'s density bound holds, and (A2) bounds its probability by $2Lq^*\delta + o(\delta)$; the excursion event is absorbed into r_K. For the last term, decompose δ^* into (a) the variance-estimation error of s^2_{plug}, which has relative standard error $\le \sqrt{(\kappa-1)/K}$ by the fourth-moment bound and, by the boundedness clause of (A3), a Bernstein tail $\Pr\big(|\widehat{s}^2_{\text{plug}}/s^2_{\text{plug}} - 1| > u\big) \le 2\exp\{-cKu^2\}$ for u below a constant; (b) the guard subtraction, whose own standard error contributes the z-term with the same tail; and (c) the bootstrap error $\max_{c,t} |\widehat{v}_c^2 - v_c^2|/s^2$, bounded by $\sqrt{c_v \log(KT)/B}$ via the Bernstein bound and a union bound over the KT cells. The $\log K$ and $\log(KT)$ factors inside $\delta_{K,B}$ are what make these exponential tails summable: choosing $\delta = \delta_{K,B}$ with C_1, C_2 large enough constants makes

$\Pr(\delta^* > \delta_{K,B}) \leq \eta_{K,B} = O(K^{-1/2})$ (polynomial-in-$1/K$ bounds of any desired order are available by inflating C_1, C_2).

(iii) The deployed scale subtracts $[\widehat{\overline{v^2}} - z\ \mathrm{SE}]_+$ rather than $\widehat{\overline{v^2}}$. On the *guard event* $\mathcal{E}_{\text{guard}} = \{[\widehat{\overline{v^2}}(t) - z\ \mathrm{SE}(t)]_+ \leq \overline{v^2}(t)\ \forall t\}$ the subtracted amount does not exceed the true design variance at any threshold, so $\widehat{s}^2_T(t) \geq \max\{\widehat{s}^2_{\text{plug}}(t) - \overline{v^2}(t), \text{floor}\}$: the subtraction component of Δ_0 errs on the wide side of the oracle scale and can only *increase* coverage, while the sampling error of $\widehat{s}^2_{\text{plug}}$ itself remains two-sided and is the part of Δ_0 that (ii) prices. This is a deterministic conditional statement. The deployed $z = 1.645$ is a normal-reference operational margin: (A3) supplies Bernstein concentration for $\widehat{\overline{v^2}}$, not an exact normal pivot, so no theorem-level probability is attached to $\mathcal{E}_{\text{guard}}$ here (a finite-sample margin with a stated level can be read off the Bernstein bound if one is wanted; and the un-Bonferronized per-threshold z means that even under a normal reference the uniform level would be $1 - T\alpha_2$, weak or vacuous at $T \approx 10$–26, with $z = \Phi^{-1}(1 - \alpha_2/T)$ the configuration under which a uniform statement is nontrivial — z is a free parameter of the shipped guard). The coverage bound in (ii) does not use (iii). □

Extension: the leave-one-out-centered deployment. Theorem 4 is stated for the independent calibration curves of (A1). The deployed high-K pipeline feeds it something else: writing Y_c for the observed departure curves, the calibration objects are the leave-one-out-centered errors $\widetilde{G}_c^{\text{loo}} = Y_c - \bar{Y}_{-c}$, and the target is scored against the transport center $\bar{Y}$. These are dependent by construction — they sum to zero identically — so (A1)'s i.i.d. structure cannot hold for them and clause (i)'s exactness does not transfer verbatim. For the *anchor* branches this same dependence is handled exactly for the survey-estimate target by Proposition S1's $K/(K-1)$ inflation; their *latent*-target validity under (A1), which Lemma S4 below establishes for raw-centered scores, is carried to the leave-one-out center by Proposition S3, which prices the centering the same way as for the deconvolution branch, whose guarantee already carries an estimation remainder: the honest resolution in both cases is to price the centering inside that remainder.

Proposition S2 (LOO-centered deconvolution). *Assume (A1)–(A3) for the raw curves, $Y_c(t) = \sqrt{s_R(t)^2 + v_c(t)^2}\, W_c(t)$, normalized to $\mathbb{E}\, W(t) = 0$ (a common location shift is removed by the centering, so this is a normalization, not a restriction), and assume the studentizer is bounded below by a constant $\omega_0 > 0$. (We state this as an assumption, not an implementation fact: the shipped floor is* relative *— the scale is floored at $0.05 \max_t \widehat{s}_{plug}(t)$ — so $\omega_0 > 0$ holds whenever the population scale is nondegenerate, $\inf_K \max_t s_{plug}(t) \geq c > 0$, which the scale family of (A1) with $s_R > 0$ supplies.) Build the band exactly as in Theorem 4(ii) but from $\widetilde{G}_c^{loo}$, the latent target scored against $\bar{Y}$. Then*

$$\Pr\left(F_{new} \in \widehat{B}\right) \;\geq\; 1 - \alpha_{dec} - \varepsilon_{K,B} - \gamma_K, \qquad \gamma_K = \frac{4LC_3}{\omega_0}\sqrt{\frac{\log\left((K{+}1)T\right)}{K}} + O(K^{-1/2}),$$

with C_3 an absolute constant from the boundedness clause of (A3). Since γ_K has the same order as $\varepsilon_{K,B}$'s K-term, we absorb it into $\varepsilon_{K,B}$ (enlarged constants) wherever the remainder is quoted, including Theorem 5's $K \geq 94$ clause.

Proof. The comparison with Theorem 4(ii)'s independent construction factors into two perturbations, one to the studentizer and one to the numerator; we chain them.

Studentizer. Write $\omega_c^{\text{raw}} = \sqrt{\widehat{s}_T^{\text{raw}\,2} + \widehat{v}_c^2}$ for the studentizer computed from the raw curves (the object Theorem 4(ii) already compares to its oracle) and ω_c^{loo} for the one computed from the centered curves — the deployed object. The centering identity is exact: $\widetilde{G}_c^{\text{loo}} = \frac{K}{K-1}(Y_c - \bar{Y})$, so the per-threshold calibration spread, which is taken about the calibration mean in both cases, satisfies $\widehat{s}_{\text{plug}}^{\text{loo}}(t) = \frac{K}{K-1}\, \widehat{s}_{\text{plug}}^{\text{raw}}(t)$ *exactly*, and the derived scale inherits a relative perturbation of at most $(\frac{K}{K-1})^2 - 1 = O(1/K)$ through the guarded subtraction and the relative floor. This is a multiplicative studentizer perturbation of exactly the form of Δ_c in the proof of Theorem 4(ii); it chains into δ^* there, enlarging $\delta_{K,B}$'s constants and nothing else.

Numerator. Now fix the studentizer at ω_c^{loo} and let S_c (resp. $\widetilde{S}_c$) be the score of the raw (resp. centered) numerator under it. Because $\max_t |\cdot|/\omega(t)$ obeys the triangle inequality, $|\widetilde{S}_c - S_c| \le \max_t |\bar{Y}_{-c}(t)|/\omega_c^{\rm loo}(t) \le \zeta/\omega_0$ for every $c \le K$, and the target's score moves by at most $\max_t |\bar{Y}(t)|/\omega_0 \le \zeta/\omega_0$, where $\zeta = \max\big(\max_c \max_t |\bar{Y}_{-c}(t)|, \max_t |\bar{Y}(t)|\big)$. Each $\bar{Y}_{-c}(t)$ and $\bar{Y}(t)$ is an average of at least $K-1$ independent, mean-zero, uniformly bounded terms ((A3)), so Hoeffding's inequality with a union bound over the $(K{+}1)T$ cells gives $\zeta \le C_3\sqrt{\log((K{+}1)T)/K} =: \gamma$ outside an event of probability $O(K^{-1/2})$. A uniform score perturbation moves order statistics by no more than itself, so $\widehat{q}^{\rm loo} \ge \widehat{q} - \gamma/\omega_0$, and a miss of the LOO band implies the independent construction's target score exceeds $\widehat{q} - 2\gamma/\omega_0$. The proof of Theorem 4(ii) applies verbatim to that event, with the additive window $(\widehat{q}{-}2\gamma/\omega_0, \widehat{q}]$ priced by (A2)'s density bound exactly as the multiplicative scale window was: probability at most $2L \cdot 2\gamma/\omega_0$ up to the $M_{(m)}$-concentration terms already inside r_K. Chaining the two perturbations — multiplicative studentizer into δ^*, additive numerator into the window — completes the proof. □

Proposition S3 (Latent validity of the leave-one-out-centered anchors)**.** *Assume (A1)–(A3) for the raw curves $Y_c(t) = \sqrt{s_R(t)^2 + v_c(t)^2}\, W_c(t)$ (normalized as in (A1)), with (A2) at the anchor level: the i.i.d. latent scores $R_c^{lat} = \max_t s_R(t)|W_c(t)|$, $c \le K{+}1$, have density at most L on an h_0-neighbourhood of their $(1-\alpha_{anchor})$ quantile q^{lat} and mass at least η on each side of it there. Build the clustered anchor exactly as deployed: leave-one-out-centered calibration errors $\widetilde{G}_c^{loo} = Y_c - \bar{Y}_{-c}$, scores $a_c = \max_t |\widetilde{G}_c^{loo}(t)|$, $\widehat{q}^{loo}$ their $\lceil(1-\alpha_{anchor})(K{+}1)\rceil$-th order statistic, and band $\bar{Y} \pm \frac{K}{K-1}\widehat{q}^{loo}$. Then*

$$\Pr\big(F_{new} \in \widehat{B}_{anchor}^{loo}\big) \;\ge\; 1 - \alpha_{anchor} - \gamma_K^{anch}, \qquad \gamma_K^{anch} = 4LC_3\sqrt{\frac{\log\big((K{+}1)T\big)}{K}} + O(K^{-1/2}),$$

and the same bound holds for the conservative anchor, whose band contains the clustered one. Since γ_K^{anch} has the order of $\varepsilon_{K,B}$'s K-term, it is absorbed into $\varepsilon_{K,B}$ wherever Theorem 5's $K \ge 94$ clause is quoted, exactly as γ_K is.

Proof. Write $m = \lceil(1-\alpha_{\rm anchor})(K{+}1)\rceil$ and, for any scores $x_1, \ldots, x_K$, $x_{(m)}$ for their m-th order statistic. The proof uses the heterogeneous observed scores $\widetilde{R}_c = \max_t |Y_c(t)|$ only through one pathwise inequality and never assumes them identically distributed.

Step 1 (domination passes to order statistics). Under (A1), $\sqrt{s_R(t)^2 + v_c(t)^2} \geq s_R(t)$ pointwise, so $\widetilde{R}_c \geq R_c^{\text{lat}}$ for every $c \leq K$ on the same W_c. Componentwise domination of two vectors is inherited by their sorted versions, hence $\widetilde{R}_{(m)} \geq R_{(m)}^{\text{lat}}$.

Step 2 (leave-one-out perturbation of the calibration scores). $|a_c - \widetilde{R}_c| = \big| \max_t |Y_c - \bar{Y}_{-c}| - \max_t |Y_c| \big| \leq \max_t |\bar{Y}_{-c}(t)| \leq \zeta$ for every $c \leq K$, with $\zeta = \max\big(\max_c \max_t |\bar{Y}_{-c}(t)|, \max_t |\bar{Y}(t)|\big)$ as in Proposition S2; a uniform perturbation of the scores moves any order statistic by at most itself, so $\widehat{q}^{\text{loo}} \geq \widetilde{R}_{(m)} - \zeta \geq R_{(m)}^{\text{lat}} - \zeta$.

Step 3 (the target). The latent target's deployed score satisfies $\max_t |G_{K+1}(t) - \bar{Y}(t)| \leq R_{K+1}^{\text{lat}} + \zeta$.

Step 4 (miss inclusion). Since $\frac{K}{K-1} \geq 1$, a miss of the deployed band implies $R_{K+1}^{\text{lat}} + \zeta > \widehat{q}^{\text{loo}} \geq R_{(m)}^{\text{lat}} - \zeta$, i.e. $R_{K+1}^{\text{lat}} > R_{(m)}^{\text{lat}} - 2\zeta$.

Step 5 (size of the perturbation). By (A3)'s boundedness and $\mathbb{E}\,W = 0$, Hoeffding's inequality with a union bound over the $(K{+}1)T$ cells gives $\zeta \leq \gamma := C_3\sqrt{\log((K{+}1)T)/K}$ outside an event of probability $O(K^{-1/2})$.

Step 6 (the latent scores are i.i.d.). On the complement,

$$\Pr\big(R_{K+1}^{\text{lat}} > R_{(m)}^{\text{lat}} - 2\gamma\big) \leq \Pr\big(R_{K+1}^{\text{lat}} > R_{(m)}^{\text{lat}}\big) + \Pr\big(R_{(m)}^{\text{lat}} - 2\gamma < R_{K+1}^{\text{lat}} \leq R_{(m)}^{\text{lat}}\big).$$

The $K{+}1$ latent scores are i.i.d. under (A1), so the first term is at most α_{anchor} by the conformal rank identity. For the second, R_{K+1}^{lat} is independent of $R_{(m)}^{\text{lat}}$, so the probability equals $\mathbb{E}\big[F_{\text{lat}}(R_{(m)}^{\text{lat}}) - F_{\text{lat}}(R_{(m)}^{\text{lat}} - 2\gamma)\big]$ with F_{lat} the common distribution function. On $\{|R_{(m)}^{\text{lat}} - q^{\text{lat}}| \leq h_0/2\}$, and for K large enough that $2\gamma \leq h_0/2$, the window lies inside the h_0-neighbourhood where (A2)'s density bound holds, so the term is at most $2L \cdot 2\gamma$; off that event, Lemma S2 applied to the i.i.d. $\{R_c^{\text{lat}}\}_{c \leq K}$ under (A2)'s mass condition gives probability $O(e^{-cK})$.

Collecting terms, $\alpha_{\text{anchor}}+4L\gamma+O(K^{-1/2})+O(e^{-cK}) = \alpha_{\text{anchor}}+\gamma_K^{\text{anch}}$. The $K/(K-1)$ inflation only enlarges the band, and the conservative band contains the clustered one, so both statements follow. (Machine check: `test_loo_validity.py::test_loo_centered_anchor_latent_cover` runs the (A1) family with country-specific v_c through the deployed centering and pins the centering cost against the raw-centered anchor on the same draws.) □

Contract test: `tests/test_loo_validity.py::test_loo_centered_deconvolution_coverage` runs the (A1) scale-family DGP *through the leave-one-out centering* — the construction the pipeline actually deploys, $\sum_c \widetilde{G}_c^{\text{loo}} = 0$ verified in-test — and checks the always-deconvolve band's latent-target coverage at activation-regime K, under a bounded-symmetric W matching (A3)'s boundedness clause literally and a Gaussian W for robustness.

Remark S4 (What (A1) rules out, and why that is the right boundary). (A1) is a distributional-matching condition: the observed and latent scores must sit in a common scale family indexed by the design variance. Remark S1's bimodal construction violates it and defeats *any* latent-target band built on observed scores, so a condition of this type is not an artifact of our proof but a feature of the problem. When (A1) is doubtful the pipeline does not rely on it: gate D (stability) and the conservative fallback do not use (A1), and at survey scale Theorem 5's guarantee bypasses this theorem entirely. Machine check: `test_estimated_law_validity.py` verifies (i)-type exactness and the K/B decay of the remainder on a scale-family DGP, and the bimodal DGP's exclusion is verified by `test_latent_target_brid`

S1.5 Theorem 5 (safe-adaptive finite-K validity—the deployed pipeline)

The deployed validity argument uses *no conditional-on-selection step*. It could not: the selection event and the conformal quantile are functions of the same calibration scores, so conditioning on the former breaks the rank-uniformity the marginal guarantee rests on. Validity instead follows from the nesting of the anchor branches, which makes selection among them free, and from a separate error budget for the deconvolution branch.

Lemma S3 (Free selection among nested bands)**.** *Let* $\widehat{B}_1 \subseteq \widehat{B}_2$ *be two bands (as random sets) with* $\Pr(F_{new} \in \widehat{B}_1) \geq 1-\alpha$*, and let* $\widehat{J} \in \{1,2\}$ *be* any *selection rule, measurable or adversarial, data-dependent or not. Then* $\Pr(F_{new} \in \widehat{B}_{\widehat{J}}) \geq 1-\alpha$*.*

This observation is elementary and belongs to the nested view of conformal prediction developed by Gupta et al. (2022); the general problem of selecting among conformal sets, including when selection is free and when it costs, is treated by Yang and Kuchibhotla (2024). We claim no novelty for the lemma itself—its role here is architectural: engineering the deployed branches so that the lemma applies (unstudentized scores make the dominance pointwise, hence the nesting) is what removes the selection problem from the validity budget.

Proof. $\{F_{\text{new}} \notin \widehat{B}_{\widehat{J}}\} \subseteq \{F_{\text{new}} \notin \widehat{B}_1\}$ pointwise: if the selected band misses, then a fortiori the narrower band $\widehat{B}_1$ misses (on $\{\widehat{J}=2\}$ because $\widehat{B}_1 \subseteq \widehat{B}_2$, on $\{\widehat{J}=1\}$ trivially). Take probabilities. □

Statement (Theorem 5). Deploy the selector of Algorithm 1 with the unstudentized PCB and conservative branches (scores $\max_t |\widetilde{G}_c(t)|$ and $\max_t(|\widetilde{G}_c(t)| + z\,\widehat{v}_c(t))$ respectively, both calibrated at level α_{anchor}) and the deconvolution branch calibrated at its own budget α_{dec}, where $(\alpha_{\text{anchor}}, \alpha_{\text{dec}}) = (\alpha, 0)$ when gate B is infeasible at the observed K (Proposition 2: $K < 94$ at the frozen constants, which covers every certification sample analyzed in this paper—ESS $K \approx 30$, LAPOP $K \approx 20$, and the WVS $\geq$ 2-wave sets at $K = 59$–77; the WVS *full item-coverage* sets at $K \approx 95$–105 are feasible in principle but are not certification samples) and otherwise $\alpha_{\text{dec}} = \min\{\max(0.1\alpha,\, 3/(K{+}1)),\, \alpha/2\}$, $\alpha_{\text{anchor}} = \alpha - \alpha_{\text{dec}}$ — a function of K only (hence frozen, not data-dependent), floored at $3/(K{+}1)$ so the branch's conformal quantile is a genuine order statistic rather than the sample maximum near the feasibility boundary. Then, finite-sample and with no conditions on the selection rule,

$$\Pr\left(\widetilde{F}_{\text{new},r}(t) \in \widehat{B}(r,t)\ \forall r,t\right) \;\geq\; 1-\alpha \quad (K < 94,\ \text{deconvolution unreachable}),$$

exactly and with exchangeability alone, for the survey-estimate target T1; the latent target

T2 carries in addition the two-term gap of Theorem 2(b), expected to be small — not assumed away: the gap is a tail-plus-anti-concentration quantity, and Remark S1 shows small symmetric noise alone does not close it — at the observed $\widehat{\rho} \leq 0.22$ (national level; at most 0.29 in the ESS subgroup scan). When $K \geq 94$, so that the deconvolution branch is reachable, then *under* (A1)–(A3),

$$\Pr\left(F_{\text{new},r}(t) \in \widehat{B}(r,t)\ \forall r,t\right) \ \geq\ 1 - \alpha - \varepsilon_{K,B},$$

now for the *latent* target directly, where $\varepsilon_{K,B}$ is the estimated-law remainder of Theorem 4 — enlarged to absorb the leave-one-out-centering terms γ_K of Proposition S2 (deconvolution branch) and γ_K^{anch} of Proposition S3 (anchors), both of the same order — and reported at deployment through the observable diagnostic $\widehat{\delta}_{\text{UCB}}(D)$. The two clauses therefore state different targets, because they rest on different assumptions: exchangeability alone reaches T1; (A1)–(A3) — which the $K \geq 94$ clause already inherits from Theorem 4 — additionally reach T2, via the following domination lemma. Without (A1)–(A3) the $K \geq 94$ worst case is $\alpha_{\text{anchor}} + \Pr(\widehat{J} = \text{dec})$ for T1.

Lemma S4 (Anchor domination under the scale family)**.** *Under (A1), let* $M_c = \max_t |W_c(t)|$ *and let* $R_{K+1}^{lat} = \max_t s_R(t)|W_{K+1}(t)|$ *be the latent target's unstudentized score. Each calibration score satisfies* $\widetilde{R}_c = \max_t \sqrt{s_R(t)^2 + v_c(t)^2}\,|W_c(t)| \ \geq\ \max_t s_R(t)|W_c(t)|$ *pointwise in t, so* $\widetilde{R}_c$ *stochastically dominates an i.i.d. copy of* R_{K+1}^{lat}*. Hence the anchor quantile* $\widehat{q}$*, an order statistic of* $\{\widetilde{R}_c\}$*, stochastically dominates the same order statistic of K i.i.d. copies of* R^{lat}*, and* $\Pr(R_{K+1}^{lat} \leq \widehat{q}) \geq \lceil (1-\alpha_{anchor})(K+1)\rceil/(K+1) \geq 1 - \alpha_{anchor}$*: the anchor branches are valid for the* latent *target, not merely conservative for it, under (A1).*

Proof. $\sqrt{s_R^2 + v_c^2} \geq s_R$ pointwise gives the pointwise inequality on the integrands, hence on the suprema, hence $\widetilde{R}_c \geq R_c^{\text{lat}}$ almost surely for the copy built from the same W_c. Coupling the K calibration units to K i.i.d. copies of R^{lat} in this way makes every order statistic of $\{\widetilde{R}_c\}$ at least the corresponding order statistic of the copies, and the conformal rank identity applied to the $K+1$ i.i.d. latent scores gives the displayed bound. $\square$

With Lemma S4 the $K \geq 94$ union bound runs on the latent target throughout: the anchor term is α_{anchor} by the lemma for raw-centered anchors and $\alpha_{\text{anchor}} + \gamma_K^{\text{anch}}$ by Proposition S3 for the deployed leave-one-out center, the deconvolution term is $\alpha_{\text{dec}} + \varepsilon_{K,B}$ by Theorem 4(ii), which is a latent-target statement to begin with. (Machine check: `test_anchor_domination.py`.) Note the lemma uses (A1) only through the common shape process; it is exactly the shape restriction that Remark S1 shows is unavoidable for *any* latent-target claim built on observed scores.

Proof. The conservative score dominates the PCB score pointwise ($|\widetilde{G}_c(t)| + z\widehat{v}_c(t) \geq |\widetilde{G}_c(t)|$, $\widehat{v} \geq 0$), so at the common level α_{anchor} the conservative order statistic is at least the PCB one and the two bands are nested: $\widehat{B}_{\text{PCB}} \subseteq \widehat{B}_{\text{con}}$. By Theorem 3 the PCB band is exact at α_{anchor}; by Lemma S3 any choice between the two nested branches retains $\Pr(\text{miss}) \leq \alpha_{\text{anchor}}$. When gate B is infeasible the selector can only ever choose between these two (Proposition 2 is a *deterministic* property of the frozen diagnostic, Section below), so the deployed guarantee is $1 - \alpha_{\text{anchor}} = 1 - \alpha$ with no remainder and no selection conditions. When gate B is feasible the deconvolution branch (not nested; narrower) is charged its own miss budget through a union bound:

$$\Pr(\text{miss}) \;=\; \Pr(\text{miss} \wedge \widehat{J} \neq \text{dec}) + \Pr(\text{miss} \wedge \widehat{J} = \text{dec}) \;\leq\; \alpha_{\text{anchor}} + \Pr(\text{miss}_{\text{dec}}) \;\leq\; \alpha_{\text{anchor}} + \alpha_{\text{dec}} + \varepsilon_{K,B},$$

the first term by the nested-selection argument restricted to $\{\widehat{J} \neq \text{dec}\}$ (the event $\{\text{miss} \wedge \widehat{J} \neq \text{dec}\}$ implies a miss of $\widehat{B}_{\text{PCB}}$), the second by Theorem 4's *marginal* guarantee for the always-deconvolve band at level α_{dec}. No term conditions on $\widehat{J}$. $\square$

Remark S5. The gates (A–D) therefore no longer carry any validity burden: they are an efficiency-and-robustness device that decides *which* of three individually budgeted constructions to return. This is weaker machinery than the earlier conditional claim and strictly stronger guarantees: at survey scale the deployed band is exact, full stop, and the only remainder anywhere in the pipeline is Theorem 4's $\varepsilon_{K,B}$, confined to the $K \geq 94$ regime where deconvolution can activate and reported through $\widehat{\delta}_{\mathrm{UCB}}(D)$ when it does.

S1.6 Proposition 2 (survey-scale unreachability)

Statement. The deployed selector activates deconvolution only if a *need* gate $\widehat{\rho}_{\mathrm{LCB}} > \rho_0$ and a *reliability* gate $D \leq \tau_D$ both hold, with $D = \max_t \widehat{\mathrm{SE}}(\widehat{s}_T^2)/\widehat{s}_T^2$ computed by the frozen formula below. Two facts bound these. *(a, algorithmic — deterministic.)* $\rho = \sqrt{\overline{v^2}}/s_{\mathrm{plug}} \in [0,1)$ with $s_{\mathrm{plug}}^2 = s_R^2 + \overline{v^2}$; and the deployed diagnostic satisfies $D \geq \sqrt{2/(K-1)}$ *identically, for every possible dataset*, so gate B cannot open unless $K \geq 1 + 2/\tau_D^2$, i.e. $K \geq 94$ at the frozen $\tau_D = 0.147$. This is a property of the frozen procedure, not a statistical estimate: the deployed pipeline provably never deconvolves at cross-national-survey K, which is what Theorem 5 uses. *(b, statistical — fourth-moment dependent.)* The interpretation of the floor as *the* sampling error of a K-cluster variance is exact under Gaussian scores and otherwise holds up to the kurtosis: the true relative SE of $\widehat{s}^2$ is $\sqrt{(\kappa-1)/K}\,(1+o(1))$ with κ the score kurtosis, larger than the Gaussian formula for heavy-tailed scores ($\kappa > 3$) and smaller for platykurtic ones ($\kappa < 3$).

Proof. (a) $\rho^2 = \overline{v^2}/(s_R^2 + \overline{v^2}) \in [0,1)$, so the need gate can fire only when the design noise $\overline{v^2}$ is an appreciable fraction of the between-population signal s_R^2. The frozen diagnostic computes $\widehat{\mathrm{SE}}(\widehat{s}_T^2) = \sqrt{2s_{\mathrm{plug}}^4/(K-1) + (\mathrm{SD}_K(v^2)/\sqrt{K})^2}$ and, by construction of the finite-K-guarded scale, $\widehat{s}_T \leq s_{\mathrm{plug}}$ for every t; hence

$$D = \max_t \frac{\widehat{\mathrm{SE}}(\widehat{s}_T^2)}{\widehat{s}_T^2} \;\geq\; \frac{\sqrt{2s_{\mathrm{plug}}^4/(K-1)}}{s_{\mathrm{plug}}^2} \;=\; \sqrt{\tfrac{2}{K-1}}$$

as an algebraic identity of the deployed statistic (checked numerically in `test_prop1_floor.py`). Hence $D \leq \tau_D$ forces $\sqrt{2/(K-1)} \leq \tau_D$, i.e. $K \geq 1 + 2/\tau_D^2$. With the frozen gate-B constants $\widehat{\delta}_{\mathrm{UCB}}(D) = a + bD$, $a = 0.0061$, $b = 0.0943$, $\delta_{\max} = 0.02$, the threshold is

$\tau_D = (\delta_{\max} - a)/b = 0.147$, giving $K \geq 1 + 2/0.147^2 \approx 94$. Repeated cross-national surveys have $K \leq 33$ (ESS) and ~ 20–40 (LAPOP, Afrobarometer, Eurobarometer), so the gate-B feasible region is empty at survey scale regardless of ρ: even if the need gate were passed, deconvolution stays blocked. (b) is the standard $\mathrm{Var}(\widehat{s}^2) = (\mu_4 - \sigma^4\,(K-3)/(K-1))/K$ expansion. □

Remark S6. Part (a) is what the validity theorem needs, and it is unconditional. Part (b) is what the *calibration* of $\widehat{\delta}_{\mathrm{UCB}}$ needs, and its fourth-moment dependence is a real limitation in one direction only: for platykurtic scores the frozen formula over-states the sampling error, so the gate is merely over-conservative (an efficiency cost); for heavy-tailed scores it under-states it, so $\widehat{\delta}_{\mathrm{UCB}}$ can under-budget the remainder — exactly the 0.02% extreme-tail failure mode the holdout exposed (Section S3), now confined by Theorem 5 to the $K \geq 94$, α_{dec}-budgeted branch. The exact threshold $K \approx 94$ depends on the frozen gate-B constants (conservatively calibrated on the development grid); a less conservative $\delta_{\max}$ would lower it, but the calibration-free statement is the floor itself. The barrier is specific to the *cross-national* application: many-unit settings (subnational areas, schools, clinics) routinely have $K \geq 94$, where the reliability gate can open.

The floor is not the selector's. Proposition 2(b) is an algorithmic identity of the *frozen* procedure. The natural objection is that 94 is an artifact of our thresholds; the following lemma removes the procedure from the statement, for the class the deployed diagnostic lives in.

Lemma S5 (Unbiased-estimation reliability floor)**.** *At the Gaussian member of the observation model, let $\widehat{V}$ be any* unbiased *estimator of a variance σ^2 built from K exchangeable population summaries with unknown mean. Then* $\mathrm{Var}(\widehat{V}) \geq 2\sigma^4/(K-1)$*: the relative standard error of any such estimator is at least $\sqrt{2/(K-1)}$. Consequently any correction that relies on an unbiased noise-variance estimate with a relative-reliability requirement τ is infeasible below $K = 1 + 2/\tau^2$, whatever the (unbiased) estimator.*

Proof. At the Gaussian member the K summaries are iid $N(\mu, \sigma^2)$, both parameters unknown. $(\bar{X}, S^2)$ is complete and sufficient; S^2 is unbiased for σ^2, hence UMVU (Lehmann–Scheffé), with $\mathrm{Var}(S^2) = 2\sigma^4/(K-1)$. No unbiased estimator has smaller variance than the UMVU one, which is the bound; the feasibility statement solves $\sqrt{2/(K-1)} \leq \tau$ for K. Proposition 2(b)'s in-procedure floor holds by construction on every realization; the lemma adds that no alternative estimator escapes the same floor at the member where spread assignments are hardest to distinguish (the $S \equiv 0$ neighbourhood of Theorem 1's construction). The frozen $\tau_D = 0.147$ is one instantiation; 94 is its intercept, $\sqrt{2/(K-1)}$ the law for this estimator class. Two scope notes. Biased, shrinkage, or prior-informed estimators are outside the lemma; a minimax version allowing bias (bounding $\inf_{\widehat{V}} \sup_{\sigma^2} \mathbb{E}[(\widehat{V} - \sigma^2)^2/\sigma^4]$) would be the fully universal statement and is left open. And the deployed diagnostic sits inside the class: the frozen SE formula is built on the unbiased K-cluster variance, so Proposition 2(b)'s floor holds on every realization by construction, not only at the Gaussian member. $\square$

Remark S7 (Three regimes and the feasibility frontier). Two structural quantities govern the design-aware correction: its maximal width gain $1 - \sqrt{1-\rho^2}$ (AW-1 below, second-order in ρ) and the unbiased-estimation reliability floor $\sqrt{2/(K-1)}$ (Lemma S5). Together they split the (K, ρ) plane into three regimes: *unnecessary* (ρ small — there is nothing worth removing), *unlearnable* (ρ large but $K < 1 + 2/\tau^2$ — the need is real and no estimator is reliable enough to certify the subtraction), and *feasible* (ρ and K both large). The deployed gates A and B are the frozen instantiations of the two quantities. the main text's Figure 2 places this paper's datasets (per-gate diagnostics in Figure SS1): the WVS full-coverage sets sit deep in the unnecessary regime ($K = 95$–105, $\widehat{\rho}_{\mathrm{LCB}} \leq 0.10$); every ESS national-unit cell sits left of the floor ($K \leq 33$, $\widehat{\rho}_{\mathrm{LCB}} \leq 0.15$); and the small-area estimand of `e54` crosses both boundaries at once ($K = 228$–287, $\widehat{\rho}_{\mathrm{LCB}}$ up to 0.52), which is exactly where the frozen selector fired. The frontier is genuinely two-dimensional — more countries at negligible noise (WVS) and more noise at few countries (ESS) both miss it — and the refinement of the unit that raises K is the same refinement that raises ρ, which is why the small-area unit reaches

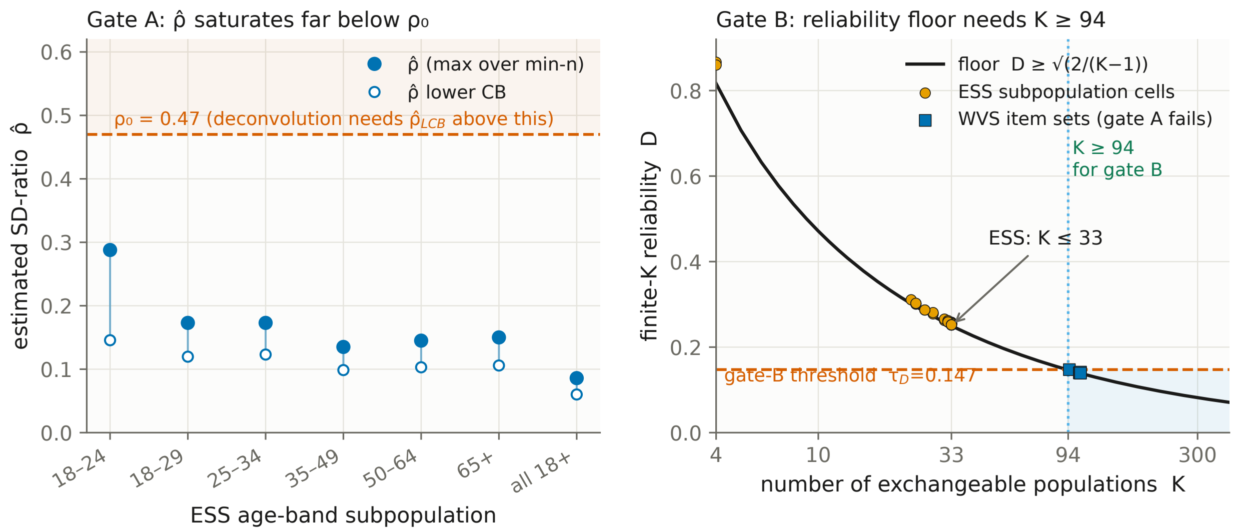

Figure S1: Per-gate diagnostics behind the main text's feasibility frontier. Left, gate A: $\widehat{\rho}$ against ρ_0 by ESS age band (each point the maximum over that band's min-n sweep, 42 cells). Right, gate B: reliability D against its floor $\sqrt{2/(K-1)}$ — the ESS sits on the floor at $K \leq 33$, and only the WVS item sets reach $K \geq 94$, where gate A fails instead.

what the national unit cannot.

S1.7 Efficiency results AW-1–AW-3 and the open oracle inequality (AW-4)

Write the three branch half-widths at level $1-\alpha$ as $W_P = q_P\, s$ (PCB), $W_D = q_D\, s\sqrt{1-\rho^2}$ (deconvolution), and $W_C = q_C\, s$ (conservative).

AW-1 (low-ρ reduction, proven). As $\rho \to 0$, $\widehat{s}_T = s\sqrt{1-\rho^2} \to s$ and the deconvolution scores converge pointwise (at fixed K) to the PCB scores, so $q_D/q_P \to 1$ by continuity of the finite-sample quantile in the studentization, and

$$W_D/W_P = (q_D/q_P)\sqrt{1-\rho^2} = 1 + o(1)\,:$$

the deconvolution band reduces continuously to PCB. The oracle scale component alone has the exact second-order expansion $\sqrt{1-\rho^2} = 1 - \frac{1}{2}\rho^2 + O(\rho^4)$; whether the full width ratio inherits it depends on the rate at which $q_D/q_P \to 1$, which the continuity argument does not quantify (a first-order term $O(\rho)$ in q_D/q_P is not excluded), so the $\frac{1}{2}\rho^2$ gain is a statement about the scale factor, not a proved bound on the band. Below ρ_0 the selector returns PCB exactly, so $W_{\text{adaptive}} = W_P$.

AW-2 (conservative dominance, conditional). For $\rho \geq \rho_0$ with stable deconvolution, $W_D/W_C = (q_D/q_C)\sqrt{1-\rho^2}$. The worst-case scores $(|E| + zv)/s$ stochastically dominate $|E|/s$, so $q_C \geq q_P$ and to first order $q_C \approx q_P(1 + z\kappa\rho)$ with $\kappa \in (0, 1]$ the correlation of the $|E|$ and v orderings across the calibration set. Hence $W_D/W_C \approx \sqrt{1-\rho^2}/(1 + z\kappa\rho) < 1$ for all $\rho > 0$, strictly and decreasing in ρ: the conservative fallback is a genuine upper band, never wider than the Bonferroni envelope and at least as wide as PCB, and deconvolution strictly narrows it whenever design noise is present. (Conditional on the stated first-order expansion of q_C.)

AW-3 (selection cannot inflate miscoverage, proven). Superseded by Theorem 5: the nested-branch lemma (any selection between $B_{\text{PCB}} \subseteq B_{\text{con}}$ is free) plus the union bound over the α_{dec}-budgeted deconvolution branch give $\Pr(\text{target} \in B_{\widehat{J}}) \geq 1 - \alpha - \varepsilon_{K,B}\mathbf{1}\{K \geq 94\}$ with *no* appeal to conditional-on-selection coverage. (The earlier mixture argument, which asserted that each branch retains its marginal guarantee conditional on $\{\widehat{J} = j\}$, was invalid: selection event and quantile share the calibration data.)

AW-4 (excess-width oracle inequality, open). Let $W^* = \min(W_P, W_D, W_C)$. Conjecture: $W_{\text{adaptive}} \leq W^* + r_{K,B}$ with $r_{K,B} = O(K^{-1/2}) + O(B^{-1/2})$ under Lipschitz width-in-ρ, the boundary zone where the target-blind ρ_0 disagrees with the oracle crossover having width $O(K^{-1/2})$ because $\widehat{\rho}$ is $\sqrt{K}$-consistent. This remains open: (i) $\rho_0 = 0.47$ is fixed from the development grid rather than derived as the oracle crossover $W_D = W_C$, and (ii) the Lipschitz constant of width-in-ρ is not established. It is stated as a candidate with a proof sketch, not a theorem.

S2 The finite-K mechanism and the development grid

Development: why the third gate is needed. On a Gaussian grid with known truth ($K \in \{30, 60, 120, 240\}$, ρ up to 1.8), the selector transitions from PCB to deconvolution at ρ_0 and to conservative at large ρ; but a naive deconvolution that omits gate B *undercovers* at small K—coverage falls to 0.75 at $K = 30$ in the transition zone—because the target scale is estimated from few countries (Figure S4). Adding the finite-K-guarded scale lifts this to 0.82, and the reliability gate B lifts the deployed worst cell to 0.862 while abstaining to conservative wherever the remainder cannot be bounded. These four transition cells at 0.862–0.878 are the development evidence that *motivates* the safety gate and *calibrates* $\widehat{\delta}_{\mathrm{UCB}}$; they are not reused below.

The finite-K mechanism, isolated. A controlled Gaussian illustration (design SD = transport SD, ρ=0.9; `fig_theory_illustration.py`) makes the source of the undercoverage explicit and shows why the deployed rule is a selector rather than a deconvolver. Both the naive and the finite-K-guarded deconvolved score scales lie *below* the oracle s_R/s_{plug} at small K (Figure S2): subtracting a design-noise variance estimated from few countries over-shrinks the scale, and the resulting band is too narrow. The safe guard shrinks less, and both converge to the oracle as $K \to \infty$. The coverage consequence is severe and grows with the trajectory length L that the band must hold simultaneously (Figure S3): naive deconvolution falls to 0.43 at K=25, L=16, and the finite-K-guarded scale improves it but does not by itself reach nominal at small K. The deployed adaptive selector detects that the remainder cannot be bounded and abstains to the conservative branch, remaining valid throughout at a width cost—the mechanism behind the K=320-only activation in the relocated confirmatory holdout below (Table S2).

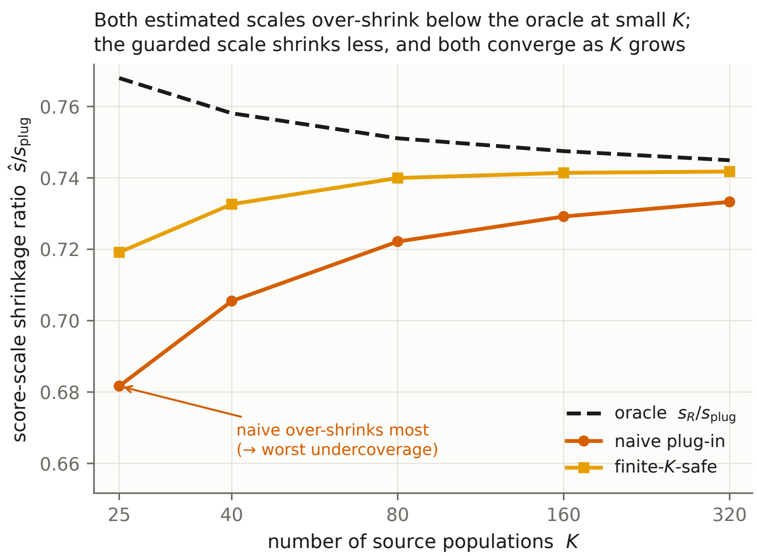

Figure S2: Score-scale shrinkage ratio $\hat{s}/s_{\text{plug}}$ vs K (Gaussian illustration, ρ=0.9, L=8). Naive and finite-K-guarded deconvolution both over-shrink below the oracle at small K—the source of the finite-K undercoverage—with the safe guard shrinking less; both converge as K grows.

Table S1: Validation provenance. No defect found in this history changed the deployed procedure: one was a harness that scored a band the package does not ship, the other a data-generating process that violated exchangeability. Both the superseded and the current runs are committed.

run	frozen before evaluation	defect found	procedure changed?	effect on reported claims
Holdout, sealed	grid and criteria (config not archived)	harness scored the S2 studentized band, not the deployed U0	no	none; it is what exposed the S2 finite-K deficit
Holdout, corrected-scorer rerun	grid rebuilt from the sealed cell table; fresh seed	—	no	450/450 cells, worst 0.882
Base-method rerun	fresh grid, ten DGPs, MC tolerance	—	band already switched S2 → U0	worst cell 0.802 → 0.893
Final sealed `e33`, run 1	script hash in `configs/`	generator normalized the target by its own SD	no — generator only	run superseded; both runs committed
Final sealed `e33`, run 2	amended generator re-hashed	—	no	120/120 cells, worst 0.878

S3 Validation of the remainder bound

What "frozen" means here, and the provenance of each run. This package uses no external study registry. Throughout, *preregistered* and *frozen* mean fixed in the replication archive before the evaluation that follows it: the analysis core, gate constants and acceptance criteria are committed in `docs/`, and the sealed runs additionally record the script hash in `configs/` before first execution. Two runs in that history exposed defects, one in a scoring harness and one in a simulation generator, and both are reported here with what they did and did not change. Table S1 is the audit trail; the rest of this section gives the numbers.

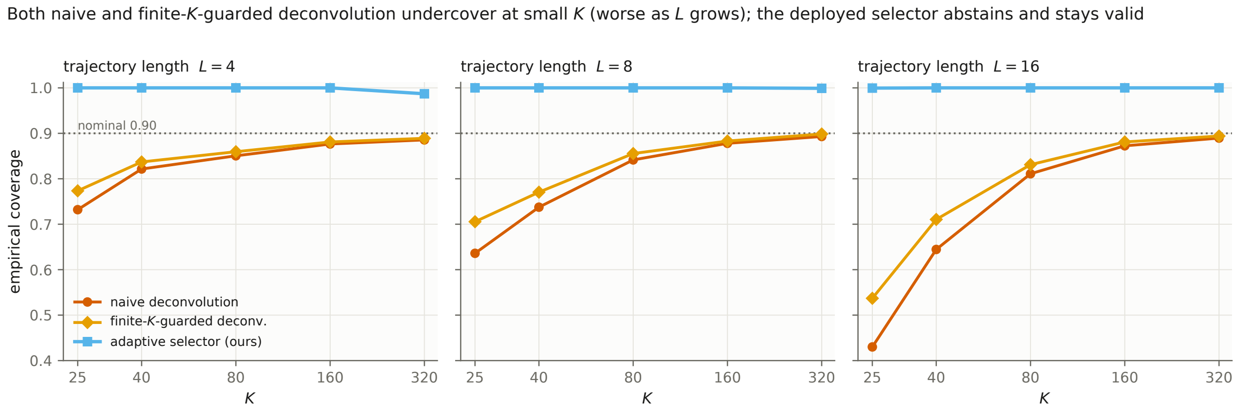

Figure S3: Latent-target coverage of the deconvolution branches vs K, faceted by trajectory length L (Gaussian illustration, ρ=0.9; nominal 0.90 dotted). Naive deconvolution undercovers severely at small K and worse as L grows; the finite-K-guarded scale improves but does not itself reach nominal at small K. The deployed adaptive selector abstains to conservative and stays valid throughout.

The failure: a finite-K deficit in the studentized base band. The sealed holdout scored the PCB branch with the S2 studentized quantiles, a different band from the U0 construction the package deploys. That is a defect in the harness, but a productive one: it is what exposed the studentized band's finite-K self-inclusion deficit, whose worst deployed cell was 0.843 (weak cross-round dependence at K=25), worst at small K and long trajectories.

The fix, and its price. The deployed band was switched to the unstudentized clustered construction, for which Theorem 3 is exact at every K. A rerun on a fresh grid ($K \in \{15, \ldots, 60\}$, ten DGPs) confirms it: the worst cell rises from 0.802 (S2, K=15) to 0.893, and every cell is compatible with its exact floor within the frozen Monte-Carlo tolerance (floor minus 2 MC-SE; 29 of 120 cells sit numerically below the floor by at most 1.7 MC-SE, as expected from binomial noise around an exact level). The price is width: 1.02–1.05× on the real ESS transport bands, since trust-CDF errors are ≈homoskedastic across the low-trust core. The cells paying more than 10% are synthetic heavy-tailed and irregular-length curves at K=15 (1.17–1.18×), where the studentized alternative is itself invalid.

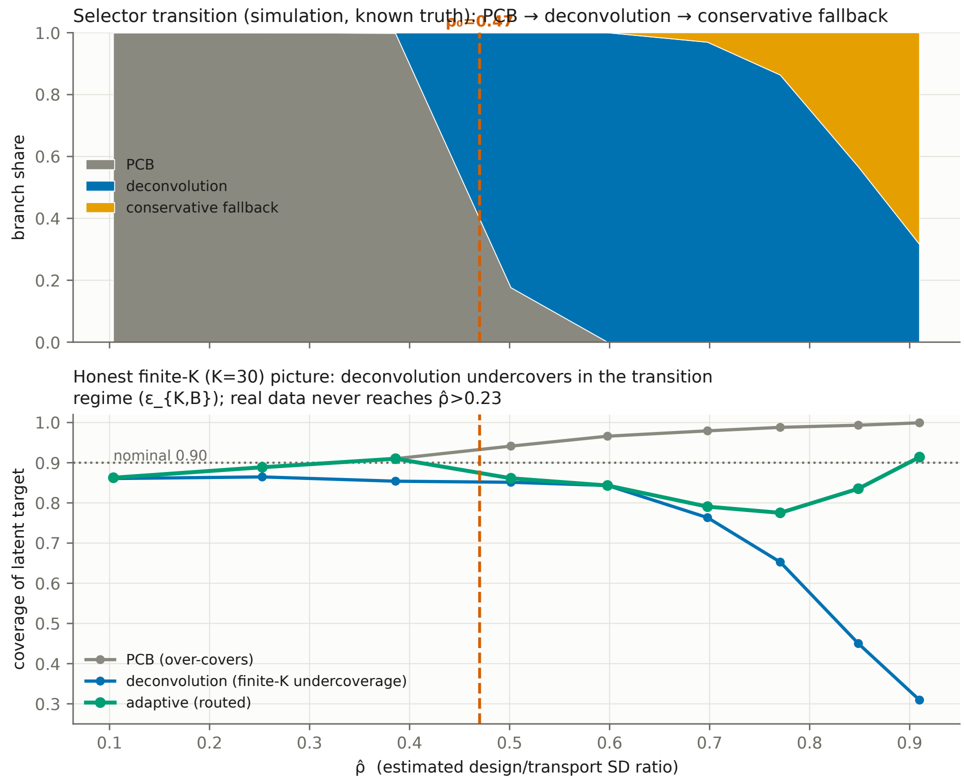

Figure S4: Development grid (Gaussian, known truth). The selector transitions PCB → deconvolution → conservative as ρ grows; naive deconvolution undercovers at small K in the transition zone—the finite-K remainder the safety gate controls.

The validation. Rerunning the same 450-cell grid on the band `dapcb` actually returns, under the α-budget architecture of Theorem 5 and a fresh seed, passes every frozen criterion outright: worst cell 0.882 (binomial-compatible with the exact level), 450/450 cells floor-compatible, deconvolution activating only at K=320, and a low-ρ width ratio of 1.005 to plain PCB. The original sealed configuration file was not preserved in the archive, so the grid is reconstructed exactly from the sealed run's cell table; both runs appear in Table S2.

A second limitation, in the extreme tail. The Gaussian-calibrated remainder bound $\widehat{\delta}_{\mathrm{UCB}}$ is not robust at the extreme: in 77 of 360,000 draws (0.02%), on strongly-dependent or heavy-tailed DGPs at K=320, draws that pass gate B undercover to ≈ 0.79. In the $\rho = 0.90$ cells the gates route 96–98% of draws to conservative and the deployed cell coverage is 0.98–0.996; at $\rho = 0.72$ the gates route far fewer draws away (deployed cell coverage 0.89–0.90

Table S2: Holdout validation of the safe selector (450 cells, 800 reps each; relocated from the main text). The *sealed* run (upper rows) scored the studentized S2 base band, whose finite-K self-inclusion deficit it exposed — the finding that motivated deploying the *unstudentized* band instead. The *corrected-scorer rerun* (lower rows) scores the band the package actually returns (U0 anchors with the α-budgeted deconvolution branch) on the same grid with a fresh seed, disclosed because the originally sealed configuration file was not preserved.

criterion	target	result	status
Sealed run, S2 scorer (historical):			
P3: design-aware never causes a sub-floor cell	sub-floor $\Rightarrow$ not deconv	**all 12 sub-0.88 cells are PCB**	pass
E2: deconvolution narrower than conservative	$\leq 0.90\times$	0.577	pass
P1: cells floor-compatible (95% MC)	all 450	442/450	diagnostic
P2: worst-cell coverage	≥ 0.86	0.843	diagnostic
Corrected-scorer rerun, deployed U0/α-budget band:			
P1: cells floor-compatible (95% MC)	all 450	**450/450**	pass
P2: worst-cell coverage	≥ 0.86	**0.882**	pass
E1: low-ρ width vs PCB	$\leq 1.05\times$	1.005	pass
E2: deconvolution narrower than conservative	$\leq 0.90\times$	0.753	pass
E3: deconvolution abstains at small K	rises with K	only K=320	pass

on the two hardest families), so the protection is concentration-dependent, not uniform over the $\rho \geq 0.72$ range. Under Theorem 5 this entire failure mode is confined to the $K \geq 94$, α_{dec}-budgeted branch and cannot touch the survey-scale guarantee.

The final sealed validation (`e33`). The main text's headline validation is a third layer: six unseen data-generating families $\times$ 120 cells, with the script hash recorded before first execution (`configs/final_validation_manifest.json`). The first sealed run failed 15 of 120 cells, all in one family (`ar2_rounds`) at every K — including K where the U0 anchor is exact for *any* exchangeable DGP. That is what diagnosed the failure as a *generator* defect rather than a pipeline one: the target trajectory was normalized by its own sample SD while calibration curves used the pooled SD, an exchangeability-violating asymmetry in the DGP itself. The generator's normalization was amended (analytic stationary SD; only that branch changed, the procedure untouched), the amended script re-hashed, and the second sealed run passes all 120 cells (worst coverage 0.878 against the 0.88 floor within Monte-Carlo tolerance, zero activations below $K = 94$). Both runs are committed (`final_validation_seal1.csv`, worst 0.53; `final_validation.csv`), and the main text's validation paragraph points here.

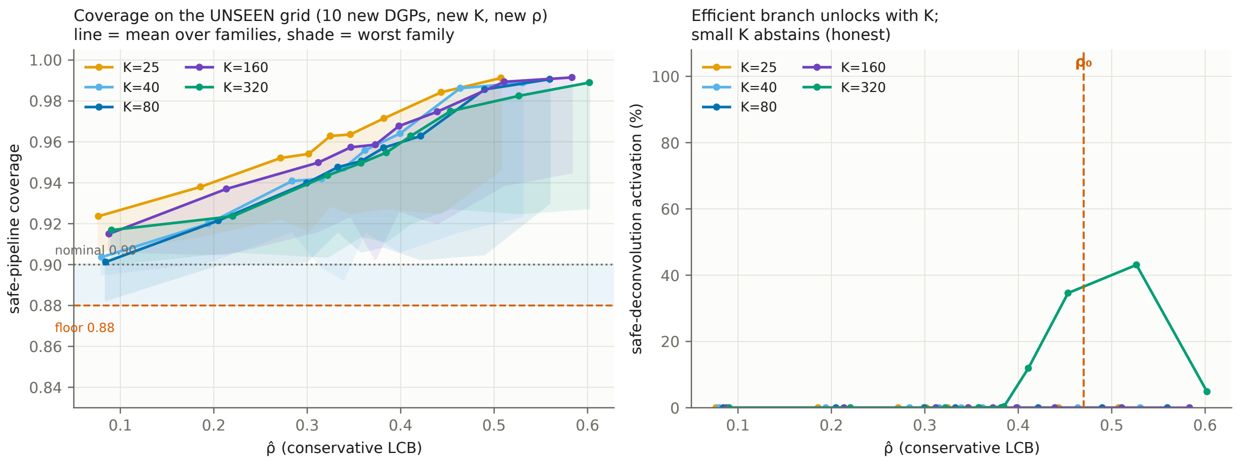

Figure S5: Holdout validation on the unseen grid (relocated from the main text). Left: deployed coverage vs. $\widehat{\rho}$ per K (line = mean over families, shade = worst family) against the 0.88 floor and 0.90 nominal. Right: deconvolution activates only at K=320; small K abstains.

S4 Additional empirical robustness

The ESS long window (rounds 1–11, 2002–2024): construction and full counts. The integrated ESS files ship usable PSU/stratum identifiers only for rounds 9–11; rounds 1–8 ship outcomes and post-stratification weights (`anweight` with `pspwght` fallback; the population-size weight is a within-country constant and drops out of country CDFs). The long-window audit (`e36`, `results/ess_long_window.csv`) therefore uses the stratified-PSU design bootstrap for rounds classified *core* (9–11) and a weights-only respondent bootstrap—the WVS treatment—for rounds classified *extended* (1–8), with the same one-directional caveat: understated design variance can only *inflate* certified counts. Pairs are adjacent ESS rounds both fielded by the country (231 pairs across 34 countries per outcome; median six pairs, maximum ten; 174 pairs have at least one weights-only side). Counts up the hierarchy, trust in parliament: any-pair 31 plug-in / 28 design-aware; net first-to-last span 14 / 9 (Cyprus, Spain, France, the United Kingdom, Greece, Hungary, Israel, Italy, Ukraine); persistent country-wide *zero* / *zero*, and zero under Bonferroni. Satisfaction with democracy: 31 / 24 any-pair, 12 / 8 net, zero persistent. Because the zero persistent count is obtained under a variance treatment that errs *inclusive*, it is robust in the disclosed direction: adding the missing design variance could only keep it at zero. Israel and Italy certify the net rung with *no* individually certified pair—their 22-year erosion is real but slower than any single wave step, the under-detection face of the wrong-unit diagnosis—while Greece certifies the net rung over its full 2002–2024 participation (rounds 1, 2, 4, 5, 10, 11), so the headline's single-pair caveat concerns the top rung only. Upgrading rounds 1–8 to the full design bootstrap requires merging the separately shipped Sample Design Data Files; that merge is delivered as `e61` ("the direct settlement" below), and changes no net or persistent count.

Same-items comparison with the Claassen latent panel (public data). Claassen's country–year support-for-democracy estimates (the corrected AJPS series; Harvard Dataverse `doi:10.7910/DVN/HWLW0J`) pool the same item families our WVS reanalysis certifies item-by-item. Merging his corrected series (change over 2006 $\rightarrow$ 2017, where his panel ends) with our $\geq$ 2-item screened universe (74 countries; `e37`): descriptive-core membership is essentially independent of the sign of his smoothed trend (6 of 13 core members fall in his series, versus 32 of 61 non-core).

That comparison is confounded two ways, as its own decomposition admits: his panel ends in 2017 while WVS wave-7 fieldwork (2017–2022) drives several of our certifications, and his latent construct excludes institutional-confidence items that drive several core memberships. Divergence produced by non-overlapping data is not information added by our method, so we removed both confounds (`e46`; `claassen_window_matched.csv`): recertifying the whole hierarchy on fieldwork ending by 2017, and restricting the comparison to the four support-for-democracy items. The independence survives matching. On the window-matched, all-item comparison (65 countries) the core declines in his series at 0.50 versus 0.49 for the rest (Fisher one-sided $p = 0.60$); on the item-matched restriction, 0.44 versus 0.50 ($p = 0.75$). So the divergence is not a window artifact and not an artifact of the institutional-trust item: what the pooled latent level and the item-level certification identify are genuinely different sets, and the certified core is not recoverable from the former. The two instruments are complements—

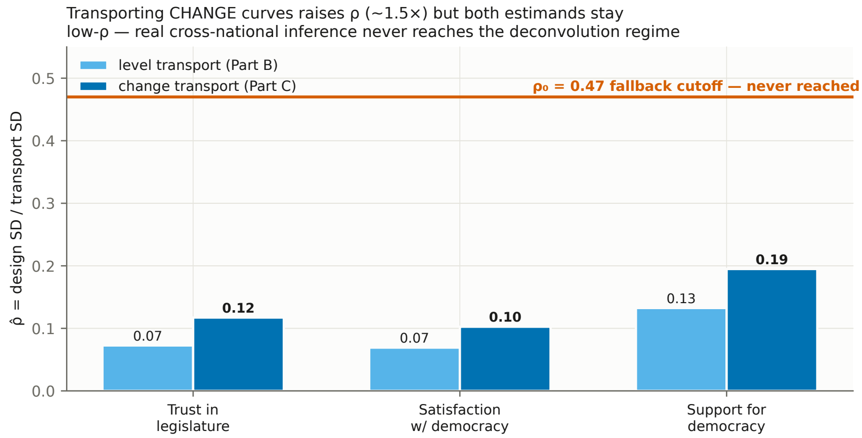

Figure S6: Design-to-total SD ratio $\widehat{\rho}$, AmericasBarometer (levels and change; $B = 200$ design bootstrap). All country-outcomes sit well below $\rho_0 = 0.47$; the estimates are downward-biased floors.

his answers a question about a smoothed latent level, ours about distribution-wide movement in a named item with finite-sample error control.

A withdrawn result, and why: the ESS-region frontier. An earlier version of this paper reported that refining the exchangeable unit to ESS NUTS regions carries $\widehat{\rho}_{\mathrm{LCB}}$ from 0.09 to 0.47, so that both barriers of the unreachability proposition become reachable on real survey data. We withdraw it. The sweep applied one estimand inconsistently: units from a country contributing a single qualifying region were left on their raw curve while all others were centred on their country's mean, and the share of un-centred units is a monotone function of the sweep variable—67% of the pool at the coarsest threshold, 34%, then 2%, then 1%. Cross-national level differences therefore inflated the denominator of ρ at the coarse end and vanished at the fine end, which is the whole of the apparent frontier. Under either consistent rule—centre every unit on its own national curve, or drop single-region countries—$\widehat{\rho}_{\mathrm{LCB}}$ is flat across the entire sweep and no crossing occurs. Three further defects compound it: ESS `region` is not one nesting level (NUTS1, NUTS2 and NUTS3 by country, with three countries entering as a single unit that *is* the country), the missing-value sentinel is treated as a region, and the "design noise" of a region is close to simple sampling error (median design effect 0.95; below one in a majority of cells). The scripts remain in the package, unused by the manuscript, so the diagnosis is reproducible.

An artifact we found and report: the apparent optimal unit. Sweeping a minimum size threshold over ESS NUTS regions produces a clean U-shape in band width, minimized at regions of ≥ 150 respondents, and we initially read it as an interior optimum in the exchangeable unit. It does not survive controls, and the reasons are instructive. First, `region` is not one nesting level: codes run three to five characters (NUTS1–3) and countries use different levels, so pooled "regions" differ in scale by an order of magnitude. Second, the threshold selects which regions qualify *and* which countries appear at all. Third and most damaging, the scored object changes along the sweep: units are taken relative to their own country's mean region curve only where that country contributes two or more qualifying regions, so at the coarse end almost every unit is a raw curve and at the fine end almost every unit is a within-country deviation—which is what drives the between-unit dispersion from 0.20 down to 0.09, not refinement.

Rebuilding the sweep without those confounds (`e52`) settles it. Within each country-round we group whole PSUs at random into units of a target size, so size is controlled rather than filtered, composition is fixed, and every unit still carries its own stratified-PSU bootstrap. Over the range where the country set is constant (targets 40 to 350; thirty countries) the half-width falls *monotonically* as units coarsen—0.190, 0.156, 0.131, 0.110, 0.093, 0.076, 0.065—with no interior minimum anywhere. Refinement is strictly costly, and the only binding constraint is the feasibility floor of the main text. We report the refuted version because the confound (an estimand that shifts silently along a sweep) is easy to reproduce and hard to see.

Two definitions of adjacency, and the persistent zero under both. The main text reports that the joint band costs France at the span rung ($9 \to 8$) and six countries at the any-pair rung ($28 \to 22$) relative to the per-rung construction, and that the two are not otherwise comparable. The reason is a definition, not a disagreement. The per-rung run (`e36`) treats a pair as adjacent when the two ESS rounds are *consecutive rounds of the survey*, so a country that skipped round 7 has no admissible pair spanning it. The joint band (`e50`) treats a pair as adjacent when the two rounds are *consecutively observed for that country*, which for a country with gaps is a coarser step and therefore a weaker requirement at the persistent rung: it asks for simultaneous decline across every observed step, not across every two-year step. The weaker definition is the one that could manufacture a persistent certification, and it does not: the count is zero under both, for both outcomes, plug-in and design-aware. The any-pair difference runs the other way, because the joint band spends one critical value on the whole contrast surface rather than a fresh one per pair, so a country whose single certified pair was marginal under the per-rung run loses it here. That is the price named in the main text, and the main text's contrast counts are computed under the joint definition throughout.

Cross-country prevalence: the closed-testing bound (`e56`). The main text's simultaneous statement — at least six of the thirty-three truly declined over their span, on each outcome — is built in two steps, both riding on machinery already in use. First, each country's joint band is inverted over its level: the span certifies at α iff the $(1-\alpha)$ quantile of the country's bootstrap sup statistic falls below $c^* = \min_t \widehat{D}(t)/\mathrm{sd}(t)$, so the smallest certifying α is the bootstrap tail probability at c^* (finite-B corrected); under a false claim the certifying event sits inside the band's own miss event, so the p-value inherits the design bootstrap's control. On both outcomes the set $\{p \leq 0.10\}$ reproduces the joint-band net set exactly (eight countries on `trstprl`, seven on `stfdem`), a check the ledger pins. Second, closed testing over the thirty-three with Simes local tests (Goeman and Solari, 2011) gives a 90% simultaneous lower confidence bound on the number of true discoveries: $d = 6$ on `trstprl` (Cyprus, the United Kingdom, Greece, Ukraine, Israel, Italy) and $d = 6$ on `stfdem` (Cyprus, Spain, Israel, Ukraine, Hungary, the United Kingdom). Simes local tests require the p-values to be independent or positively dependent (PRDS) across countries. That is a *design* assumption — conditional on the finite populations, the countries' sampling mechanisms are independent, which the ESS's country-by-country sample designs support — and not a consequence of running separate bootstrap streams. Because the assumption is not testable from the data, the same closed testing is rerun with Bonferroni local tests, which are valid under arbitrary dependence: the largest subset surviving its Bonferroni test has size $h_B = \max\{k : p_{(m-k+1)} > \alpha/k\}$, $d = m - h_B$, and the subset bound is computed by direct closure (`pcb.inference.prevalence`, pinned to exhaustive closed testing on small families in `tests/test_prevalence_local_tests.py`). The Bonferroni bound is also $d = 6$ on both outcomes and names the same six countries: the four p-values at the bootstrap floor (0.0005) and the next two (≤ 0.0025 on trust, ≤ 0.0015 on satisfaction) clear $\alpha/33$ with room, so the statement does not lean on the dependence assumption. Nor on the design layer of the early rounds: rounds 1–8 enter the shipped p-values through the weights-only respondent bootstrap of the integrated files, and the Sample Design Data File merge (`e61`, 88 country-rounds upgraded to a stratified-PSU bootstrap) reruns the p-value inversion with the same seeds (`results/ess_prevalence_sddf.csv`). Ten of the sixty-six p-values move, by at most 0.016, and the bound is unchanged under the rounds 1–8 SDDF upgrade: $d = 6$ on both outcomes under Simes and under Bonferroni local tests, naming the same six countries. Closed testing trims the certified sets' marginal members (Spain and Hungary on `trstprl`, France on `stfdem`), which is the honest gap between a per-country-α count and a simultaneous one. Naming is licensed by simultaneity over subsets, read correctly: the global d says "at least six among all thirty-three," not "these six," so the named set is re-certified on its own subset bound *within the full closed testing* — with h the largest subset of all thirty-three surviving its own Simes test, $d(S) = \max_{1 \leq u \leq |S|} [\, 1 - u + \#\{p_i \in S : p_i \leq u\alpha/h\} \,]$ (the closure shortcut of Goeman and Solari, 2011) — and on both outcomes the six smallest-p countries satisfy $d(S) = 6$, so the named sets above are themselves simultaneous 90% statements. (Running the subset as its own family would ignore the closure and can over-count; the contract tests pin the distinction and verify validity on planted truth, `tests/test_prevalence.py`.)

A valid cross-item WVS claim (`e63`, `e65`). The main text's thirteen-country $\geq$ 2-item core first counts two separate $\alpha = 0.10$ item-level certifications; that count alone does not test that two declines are true. We therefore inverted the unchanged persistent band for each available country–item and tested the partial-conjunction alternative that at least two item nulls are false. For a country with m available items, the primary arbitrary-dependence-valid value is $p_{\mathrm{PC}} = \min\{1, (m-1)p_{(2)}\}$ (Benjamini and Heller, 2008). Twelve of the thirteen descriptive-core countries pass at 0.10; Trinidad and Tobago does not ($p_{\mathrm{PC}} = 0.144$). The claim is simultaneous across items within each country, not across the twelve countries. Closing the 78 country-level p_{PC} values with Bonferroni local tests gives a simultaneous lower bound of one true country and re-certifies Rwanda by name. Inflating the weights-only bootstrap variance by 1.5 and 2.0 leaves ten and nine country-level partial-conjunction certifications; the corresponding across-country lower bounds are one and zero. The machine checks reproduce all five original E26 certification sets before combining them.

The small-area activation, in full (`e54`, `e55`). The one place in this paper where the design-aware branch fires on real data is the ESS regional estimand, and because it is the paper's only positive activation we give the run in full rather than by summary.

Estimand and design variance. For region g of country c the object is the departure $D_g = F_g - F_{\mathrm{nation}(c)}$, fixed once and applied to every unit. A single stratified-PSU bootstrap of the *whole country* recomputes F_g and F_{nation} on each of $B = 400$ replicates, so $\widehat{v}_g$ is the design SD of the difference and the induced covariance between a region and its own national curve is handled without modelling. The missing-value sentinel (`99999`) is excluded, as are the three countries whose `region` code equals the country code; a country must contribute at least two usable regions, since one region cannot express a departure. The bootstrap fallback that would fire if a stratum lost a region entirely fires in 0 of 5,500 replicate-draws.

What activates. Across 23 unit-rounds the need gate opens in 8 and the frozen selector routes to deconvolution in 4, all at $K = 228$–287 with minimum unit sizes of forty and sixty respondents in rounds 9 and 10; in the activating cells the deconvolved target scale itself sits 15–17% below its plug-in value. Round 11 never activates: its regional design noise is lower and gate A does not open at any size. Restricted to the fourteen countries whose region codes sit at a common NUTS level, gate A still opens in nine of twenty-one unit-rounds but $K \leq 140$ keeps gate B shut, so reaching the activating scale requires pooling levels—a scope condition, stated here and in the main text rather than buried.

Is the region the exchangeable unit? The regions nest in twenty-seven countries, nine apiece, and between-country differences carry 32–49% of the variance in a region's sup-departure. If the country were the exchangeable unit the effective K would be about 27, below this paper's own $K \geq 94$ floor, and the activation would be an instance of the wrong-unit error this paper diagnoses. `e55` settles it by holding out whole countries. Two schemes are compared on the same cells: leave-one-*region*-out, which keeps the held-out region's country-mates in the calibration set, and leave-one-*country*-out, which removes them all. The aggregate LOCO level is not by itself decisive—the held-out sets union to the whole pool, so it is partly self-fulfilling—and the informative statistic is the gap between the schemes, which is what within-country dependence would inflate. It is at most 0.015 across all 12 cells; the plug-in anchor's realized coverage is 0.893–0.903 against its 0.900 target. Deleting nine country-mates from the calibration set costs nothing — evidence against within-country dependence deflating the quantile, read as a diagnostic that bounds the concern rather than removes it: the guarantee remains marginal over regions and presumes their exchangeability.

What does not transfer. Coverage *conditional* on the held-out country is far from nominal: the median country is covered on all its regions, the tenth percentile is 0.75, and the three worst are Bulgaria, Greece and Hungary at 0.53–0.63—which are, in that order, among the countries with the largest regional heterogeneity. This is the ordinary marginal-versus-conditional distinction of conformal inference and we do not read the band per country. In the activating cells the selected band covers 0.93–0.98 against a nominal 0.88; that check scores the *observed* departure, while the deconvolution branch targets the latent one. The latent coverage is not directly measurable, and reading the observed-departure check as conservative for it requires the symmetric-noise and scale-family structure of (A1). We therefore read the activation as establishing feasibility — the frozen selector enters its correction regime at this unit — rather than as a direct latent-target validation.

Promised reruns, delivered. Four robustness reruns flagged in earlier drafts are now part of the package. *(a) Rao–Wu–Yue rescaled bootstrap* (`e38`; `*_rescaled.csv`): flipping every stratified-bootstrap site from m-of-m to the rescaled $m-1$ draw leaves *every* ESS certification count unchanged at every rung (12/6/1 trust, 14/8/1 satisfaction, Bonferroni included), moves the ESS subgroup-scan maximum $\widehat{\rho}$ from 0.288 to 0.281 and the LAPOP change-transport maximum from 0.212 to 0.222—both still severalfold below $\rho_0 = 0.47$—and opens zero gates (42/42 cells still route to PCB). Single-PSU strata, the one thing rescaling cannot repair, turn out to be negligible here: 1 of 7,028 strata in the core rounds, covering eleven respondents. The small-area activation is the one place the rescaling moves a decision (`e62`; `small_area_transport_rescaled.csv`): the higher rescaled design variance opens the need gate more broadly ($8 \to 13$ of 23 unit-rounds) but pushes boundary cells past the frozen reliability and efficiency thresholds, so the deployed selector fires in 1 cell rather than 4 (min-$n = 40$, round 9, $K = 273$, certified gain 0.24). The two-barrier structure and the feasibility reading are unchanged — activation persists, national-scale abstention persists — but the four-cell activation table of `e54` is specific to the shipped m-of-m calibration, which is one reason the rescaling is reported as a sensitivity rather than promoted to the deployed default. The more consequential structural fact is different and we state it plainly: in 52 of the 90 core country-rounds the median PSU contains a single respondent—these are register samples of individuals, where `psu` carries no clustering to absorb. the main text's "uniformly low design-noise regime" is therefore in substantial part a statement about which countries sample individuals from population registers, not a universal property of survey inference. *(b) Round-10 mode audit* (`e40`; `ess_mode_table.csv`), built from the data's own `mode` variable rather than from external documentation: nine countries (AT, CY, DE, ES, IL, LV, PL, RS, SE) switched to self-completion at round 10; rounds 9 and 11 are face-to-face throughout the integrated file. One further mode change the binary partition hides: video interviewing appears in twenty countries at rounds 10–11, at shares up to 0.48 (Norway), including Estonia and the Netherlands, two of the six net-certified countries in the rounds-9–11 result. We collapse video into face-to-face because no parallel-mode randomization exists in this extract to identify its effect; the certifications spanning a video introduction are conditional on that collapse, and `ess_mode_table.csv` ships the per-cell shares so a reader can exclude them. Excluding the mode-confounded pairs: the net counts are *unchanged* (6 and 8; the $9 \to 11$ endpoints are both face-to-face), the persistent count is unchanged (Greece—itself face-to-face in both certified rounds), and only the nine switchers' any-pair claims are affected (marked mode-confounded rather than re-certified). *(c) Leave-one-region-out exchangeability audit* (`e41`; `loro_exchangeability.csv`): calibrating the U0 transport band on all countries outside a region and scoring the held-out region's trajectories, Nordic/Western/Southern blocs cover at 6–8 of 6–8 (nominal 0.90), the full-pool leave-one-out baseline at 28/30 for both outcomes, but the post-communist East covers only 6/9 (trust) and 5/9 (satisfaction): a quantified scope condition—transport *into* the Eastern bloc from a non-Eastern pool is anti-conservative—consistent with the East-dominated certified core of the WVS reanalysis, and irrelevant to the within-country certifications, which use no exchangeability. *(d) Real-data severity injection* (`e42`; `real_severity.csv`). Two designs, because they answer different questions. *Power (null-imposed):* the estimand is *replaced* by an injected decline of δ CDF points over the preregistered core — $\widehat{D} \leftarrow \delta$ and the bootstrap draws recentred on it — keeping each of the 30 countries' 57 real stratified-PSU bootstraps. At $\delta = 0$ this is the sharp null and realized size is 0.001–0.007 across the three rungs (nominal 0.10); 80% power arrives at $\delta \approx 0.033$ per pair for the net rung, ≈ 0.06 for the pairwise rung, and ≈ 0.075 for the persistent rung. This reproduces E32's simulated ordering against genuine design noise. *Detection (additive):* the injection is added on top of what each country actually did, so $\delta = 0$ returns the observed certification rates (0.20 net, 0.03 persistent) and the curve answers "how much further decline would bring the certified set to a given size" — a different quantity, reported alongside. A caution for anyone reusing this design: the shift must travel with the bootstrap distribution. Adding it only to the point estimate lowers the studentized critical value by δ/sd *while* raising the margin, counting the injection twice; an earlier version of this script did so and overstated power by roughly a factor of two in δ. *(e) WVS design-effect sensitivity* (`e39`; `wvs_deff_sensitivity.csv`): rerunning the full five-item certification with the weights-only bootstrap variance inflated by deff 1.5 and 2.0: at $\times 1.5$ every per-item persistent count is within 2 of the base and the ≥ 2-item certified core is *identical* (13 countries); at $\times 2.0$ the core loses only Trinidad and Tobago (12 remain) and the marginal-to-persistent gap widens from 2.6–6.5$\times$ to 3.0–8.3$\times$ — confirming the paper's directional claim that the shipped gap is a lower bound. The stricter partial-conjunction core (`e65`) moves from 12 at baseline to 10 and 9 under the same two multipliers.

The long window, stress-tested and then settled. The rounds-1–11 analysis (`e36`) mixes design-bootstrap cells (rounds 9–11) with weights-only cells (rounds 1–8): 174 of 231 pairs have a weights-only side. Because each cell is self-studentized, the sup-t critical value is scale-free per cell and the understatement enters only through the final margins, in the inclusive direction — protective for the *zero persistent* result, a liability for the *nine net* result. Four checks bound it, and a fifth settles it directly. *(0) A scope correction.* The design-effect sensitivity below was run against the per-family construction that the joint band of the main text replaces, so it certifies the *nine*-country set rather than the eight the manuscript now reports. Rerun against the joint band, the eight-country set is stable under weights-only inflation up to a design effect of roughly 1.3; Spain, the marginal member, loses certification above that. We state the narrower interval rather than carry the old one over.

(i) Design-effect sensitivity (`e44`; `ess_long_window_deff.csv`): inflating the weights-only cells' bootstrap variance by deff 1.5 and 2.0 leaves the net counts *exactly* unchanged (9 for trust, 8 for satisfaction, same membership at every deff) and the persistent counts at zero; only the any-pair rung moves ($28 \to 25$, $24 \to 22$). The nine are not an artifact of the missing clustering variance. *(ii) Endpoint sensitivity* (`e43`; `ess_endpoint_sensitivity.csv`) re-certified the span rung over every admissible sub-span as a separate test each, and found only Cyprus, Spain and the United Kingdom surviving the removal of either endpoint. That analysis is *superseded* by the joint construction of the main text (the claim-family transfer proposition of the main text, `e50`), under which every endpoint choice is certified on one coverage event and no selection cost arises; we keep the run in the package because it shows what the per-family construction cost. *(iii) Certified magnitudes and recoveries* (`e45`, superseded by `e50`): the same one-sided band that certifies a decline returns its size, the largest d with $\widehat{D}(t) - c\,\mathrm{sd}(t) \geq d$ at every core threshold, and reversing the inequality certifies rises. Computed per family, these gave 56 rising pairs against 59 falling and 24 countries with both, at α per family; computed inside the joint band they give 23 of thirty-three at α in total, which is the number the main text reports. *(iv) The direct settlement* (`e61`; `ess_long_window_sddf.csv`, `ess_joint_claims_sddf.csv`): merging the separately distributed ESS Sample Design Data Files for rounds 1–8 (`pcb.data.ess_sddf`; per-country files for rounds 1–6, integrated files for rounds 7–8) supplies psu/stratum for 245,897 respondents and upgrades 88 country-rounds from the weights-only to the stratified-PSU design bootstrap, cutting the weights-only pairs from 174 to about 106 of 231 per outcome; country-rounds whose SDDF is not published or carries no PSU remain weights-only. Rerun with the shipped seeds so that core-round draws are identical, the upgrade changes *nothing* the manuscript reports: the joint-band net sets are unchanged with identical membership (8 trust, 7 satisfaction), persistent stays zero on both outcomes, the both-directions count stays 23, and the per-family net and Bonferroni counts are unchanged — the only movement is one marginal any-pair certification lost per outcome (Russia on trust in the joint

band, Switzerland on satisfaction in the per-family run), exactly the inclusive direction the understatement argument predicts.

Nonresponse and sample-type limits the machinery does not reach. Two exposures deserve naming rather than absorption into "measurement constancy." ESS response rates fell markedly across the 2002–2024 window, and nonresponse correlates with political trust; post-stratification on demographic margins cannot absorb a secular composition drift, which would produce exactly the slow monotone CDF movement the net rung certifies. This is a bias, not a variance, and no design bootstrap addresses it; the erosion index—the share of a country's ordered contrasts that certify—is the reading least exposed to it, since no single wave carries it. Second, the WVS trend file pools country-waves of heterogeneous sample type, including waves whose frames or substitution rules make design-based variance ill-defined; for those, "weights-only bootstrap × deff" is a sensitivity, not a variance, and the certified-core entries resting on them should be read with the per-country technical records to hand.

Design-preserving stress test, reinterpreted. Subsampling the real AmericasBarometer while preserving its stratified-PSU structure, pooled $\widehat{\rho}$ rises to ≈ 0.23 and then *turns over* at the deepest subsampling ($f = 1/16$), where the protocol's own $1/\sqrt{f}$ scaling predicts $\widehat{\rho} \approx 0.49$—i.e., a crossing of ρ_0. The discrepancy is diagnostic: at $f = 1/16$ roughly one PSU remains per stratum, where the m-of-m bootstrap (no Rao–Wu–Yue rescaling; singleton strata contributing zero) collapses the design-variance estimate exactly as the true design noise peaks, while s_{plug} continues to absorb the realized noise. What the sweep demonstrates is therefore a failure mode of the $\widehat{\rho}$ diagnostic under sparse-PSU designs—with the deployed band's coverage remaining at or above nominal throughout—not a substantive saturation law, and we have removed "saturation" claims resting on it from the main text. The rescaled-bootstrap reruns (`psu_bootstrap(rescale=True)`) are delivered as `e38` and `e62`, reported under "promised reruns" above.

The positive regime, demonstrated (E31). The scope result says where the correction is unreachable; a companion simulation shows where it pays. On the MRP-style small-area DGP (latent area curves plus known heteroskedastic posterior SDs), the frozen pipeline — unchanged constants, the α-budget architecture of Theorem 5 — is run at $K \in \{60, 94, 150, 220, 300\}$ areas and posterior-noise ratios 0.3–0.9, scoring the returned band against the latent area curve (800 reps per cell). Three facts emerge. First, gate-B *feasibility* ($K \geq 94$) is necessary but not sufficient: with the deconvolution branch honestly budgeted at $\alpha_{\text{dec}} = \max(0.1\alpha, 3/(K+1))$, its conformal quantile is informative — and the width-gain gate clears — only from $K \approx 220$, so the practical activation frontier sits near $K \approx 200$–300. Second, where it activates (noise ratio ≈ 0.7: 31% of draws at K=220, 98% at K=300) it pays: the deployed band is 0.69–0.74$\times$ the conservative envelope's width at coverage 0.98–1.00, against the guarantee floor 0.88. Third, at noise ratios near or above the between-area signal the stability gate abstains to conservative — correctly, since there the deconvolved scale hits its floor. Many-unit clustered settings (US counties, schools, clinics) have K well beyond this frontier; repeated cross-national surveys never do (`e31_positive_regime`; `results/positive_regime.csv`).

The barriers are not survey artifacts. Both gates are distribution-free, so the boundary should transfer to any conformal procedure calibrated on estimated objects (the general remark of the main text). A simulation confirms it outside the survey setting (`e29_beyond_surveys`): across Gaussian, heavy-tailed, skewed, and a non-survey model-based small-area (MRP-style) calibration, the reliability floor $D \geq \sqrt{2/(K-1)}$ holds in every case (minimum D/floor ≥ 1.06), and the gate opens only at $K \approx 130$–150—if anything a stronger barrier than the floor-implied $K \geq 94$. The one substantive difference is instructive: when a small area's posterior noise is comparable to the between-area signal ($\widehat{\rho}_{\text{LCB}} = 0.54$ at a noise-to-signal ratio of 0.5), the *need* gate does fire, unlike any cross-national survey. A many-area MRP application with appreciable posterior uncertainty is therefore the setting where the deconvolution could actually activate—the positive-result frontier the survey-scale impossibility leaves open.

The AmericasBarometer's design metadata, alongside the ESS's. The ESS publishes sample-design files (PSU, stratum, inclusion probabilities) for all rounds—integrated into the main file from round 9, as separate SDDF files earlier—so our restriction to rounds 9–11 reflects the integrated extract we analyzed, not an ESS limitation (an earlier draft misstated this); the AmericasBarometer releases full stratified-PSU structure in its merged file, letting us compute a proper design bootstrap directly. The design effect is real: the ratio of the proper stratified-PSU standard deviation to a naive independent-resampling one has median ≈ 1.14–1.21 and reaches 1.97 in the most clustered country-years. The proper band captures clustering variance the naive bootstrap misses. Yet certification is coarse-robust: the design effect moves band *width* but rarely flips the binary decline decision, because the decision is dominated by the between-country transport spread, not the within-country design noise—the same low-ρ fact, seen from the certification side.

Robustness across age groups (ESS). A preregistered age-group analysis (preregistered in the replication package) reruns the same certification within fixed age bands, changing nothing else. The "only Greece" result is not an artifact of pooling ages: it holds within the older (50+) group for both outcomes and within the middle (30–49) group for trust in parliament, so the persistent decline is concentrated in the older and middle adult population rather than created by averaging. Youth (18–29) certify *zero* persistent declines, but we read this as an underpowered null rather than youth stability: only 17 of 30 countries clear the preregistered minimum-sample gate for a single youth band, and the smaller, noisier youth cells yield wider design-aware bands that certify less—the small-K scope condition of the simulation appearing on real data. The analysis thus supports the headline while adding an honest caveat, and it is emphatically *not* a story of youth-led democratic disaffection.

Youth deconsolidation on the World Values Survey. The youth claim in the Foa–Mounk reanalysis is only half-supported: the young show *more* persistent decline in rating democracy essential (7 vs. 3 countries) but *not* in openness to strongman or army rule (0–1 vs. 4), against the "youth embrace authoritarianism" reading.

Two flagged readings in the certified core. *Finland and Switzerland.* Persistent rising openness to strongman or army rule in two of the most stable democracies contradicts most published accounts. These two items carry documented translation and response-option instabilities, and both countries' four-decade series span changes of fieldwork agency and mode—a distribution-wide shift is exactly what an administration change produces, and our machinery corrects sampling error only. We therefore report the pair as a candidate finding conditional on a within-programme measurement audit, which requires the programme-level microdata. *Turkey, the Philippines, Mexico.* These certify only on importance of democracy, an item that enters the battery in the mid-2000s; "only on `imp_dem`" therefore mixes substance with differential window length, and the reading—persistent downgrading of democracy's importance without wholesale attitudinal collapse—should be held loosely. Wave coverage varies by country and item throughout, so multi-item certification is mechanically harder where fewer items are measured.

V-Dem cross-tabs of the certified core (public data; descriptive). Two checks against the V-Dem country–year data (v15; Regimes-of-the-World classification and electoral-democracy index; `e35`, results in `vdem_regime_crosstab.csv` and `vdem_predictive.csv`). *Regime type.* Of the 13 descriptive-core countries, five (Azerbaijan, Iraq, Lebanon, Rwanda, Uzbekistan) are *electoral autocracies* and none is a closed autocracy in the 2019 classification; five are electoral democracies (Albania, Bosnia and Herzegovina, Ecuador, Ghana, Tunisia) and three liberal democracies (Finland, Switzerland, Trinidad and Tobago). The partial-conjunction screen removes Trinidad and Tobago, leaving the same five autocracies, five electoral democracies, and two liberal democracies. The main text's caution—that in the autocratic subset falling stated support reads as autocratic legitimation rather than deconsolidation in the consolidated-democracy sense—therefore applies to five of twelve partial-conjunction members, not three as a name-by-name reading suggests. *Predictive validity, a null.* Among the 76 countries screened on ≥ 2 items, certified-core membership does *not* anticipate subsequent electoral-democracy decline (change in `v2x_polyarchy`, 2022 $\rightarrow$ 2025 and 2017 $\rightarrow$ 2025) better than the marginal any-pair flag: median changes are similar (-0.014 vs. -0.009 over 2022 $\rightarrow$ 2025), one-sided Mann–Whitney $p = 0.80$, Fisher $p = 0.89$. Persistent *attitudinal* deconsolidation and near-term *institutional* backsliding are, in this window, distinct phenomena— consistent with the contested attitudes-to-institutions mechanism in the recent panel literature cited in the main text, and a bound on how far the certified core should be read as an early-warning indicator. Both cross-tabs are descriptive aggregation on small N, reported as produced.

S5 Extended related work

Our work sits at the intersection of four literatures. First, **distribution-free conformal prediction** gives finite-sample coverage without modeling assumptions (Vovk et al., 2005; Lei et al., 2018). Because political data rarely satisfy exchangeability, we build on its relaxations: weighted conformal prediction for covariate shift (Tibshirani et al., 2019) and nonexchangeable conformal prediction, which bounds the coverage gap by the total variation between exchanged distributions (Barber et al., 2023). Gibbs and Candès (2021); Gibbs et al. (2024) adapt conformal sets to distribution shift and interpolate between marginal and conditional validity. Our "adaptive" selector differs in kind: it adapts not to temporal shift but to the *certified magnitude of survey design noise*, reducing to a fixed clustered method when that noise is negligible. Within the discipline, Randahl et al. (2026) bring conformal prediction to *Political Analysis* for the first time, calibrating bin-conditional intervals around a point forecast of conflict fatalities; our object is different in kind—finite-sample *simulta-*

neous trajectory bands with the country as the exchangeable unit and estimated calibration distributions, rather than bin-conditional intervals on a single prediction.

Second, we treat the **population (country) as the exchangeable unit**, following hierarchical conformal prediction for two-layer models (Dunn et al., 2023) and distribution-free inference with hierarchical data (Lee et al., 2026). The same finite-sample-exact clustered construction is developed, for a different (non-survey) domain, in a companion paper (Park, 2026). These methods transport guarantees to an unseen cluster—our cross-national setting—but treat each cluster's data as cleanly sampled. Our departure is that the calibration objects are not observed: each country's attitude distribution is *estimated* under a complex survey design, so design-based sampling variance (Binder, 1983; Lumley, 2010) contaminates the very scores conformal calibration relies upon. The nested view of conformal prediction and selection among conformal sets are developed by Gupta et al. (2022) and Yang and Kuchibhotla (2024); our Lemma S3 is an instance of that machinery, and our contribution is architectural (engineering the branches so the lemma applies), not the lemma itself.

Third, the closest methodological neighbors handle **noisy or estimated calibration**. Wieczorek (2023) develops design-based conformal prediction under complex sampling, but design enters through test-point weighting rather than as noise in transported objects. Einbinder et al. (2024) and Sesia et al. (2025) correct conformal calibration under label noise, the latter by deconvolving a *known* noise model. Our central theoretical result is that this identification premise generally fails for survey design noise: recovering the latent transport-error law is a deconvolution problem (Fan, 1991) that is non-identified without a design-noise law, so no band built on the observed curves can beat the conservative plug-in by more than the discreteness slack while remaining valid. And even when a noise model *is* supplied, the deconvolution's reliability is itself bounded at a finite number of calibration units by an algorithmic floor—a finite-K obstruction distinct from the asymptotic identifiability the classical deconvolution literature studies.

Fourth, our target is a **functional object**—a CDF over thresholds tracked across rounds—placing us near simultaneous functional conformal bands (Diquigiovanni et al., 2022). Procedures like FDR or Bonferroni (Benjamini and Hochberg, 1995) adjust a collection of tests after the fact; our guarantee is instead carried by the geometry of a single finite-sample band, so the "persistent" object is *defined and covered directly*, not recovered by correcting marginal p-values. The distinction is one of estimand, not merely error rate, and partial-pooling cautions about reading marginal country trends (Gelman et al., 2012) motivate but do not deliver the stronger claim. Claassen-style latent-variable panels (Claassen, 2019, 2020) deliver model-based credible intervals for a latent mood under a dynamic measurement model, pooling across items and years; our bands deliver finite-sample, model-free coverage for a distribution-level object at a stated claim rung, at the price of wider intervals and no pooling. The two are complements—a certification that survives our rung hierarchy and falls inside a Claassen-style interval is doubly supported—and a same-data comparison is a natural companion exercise.

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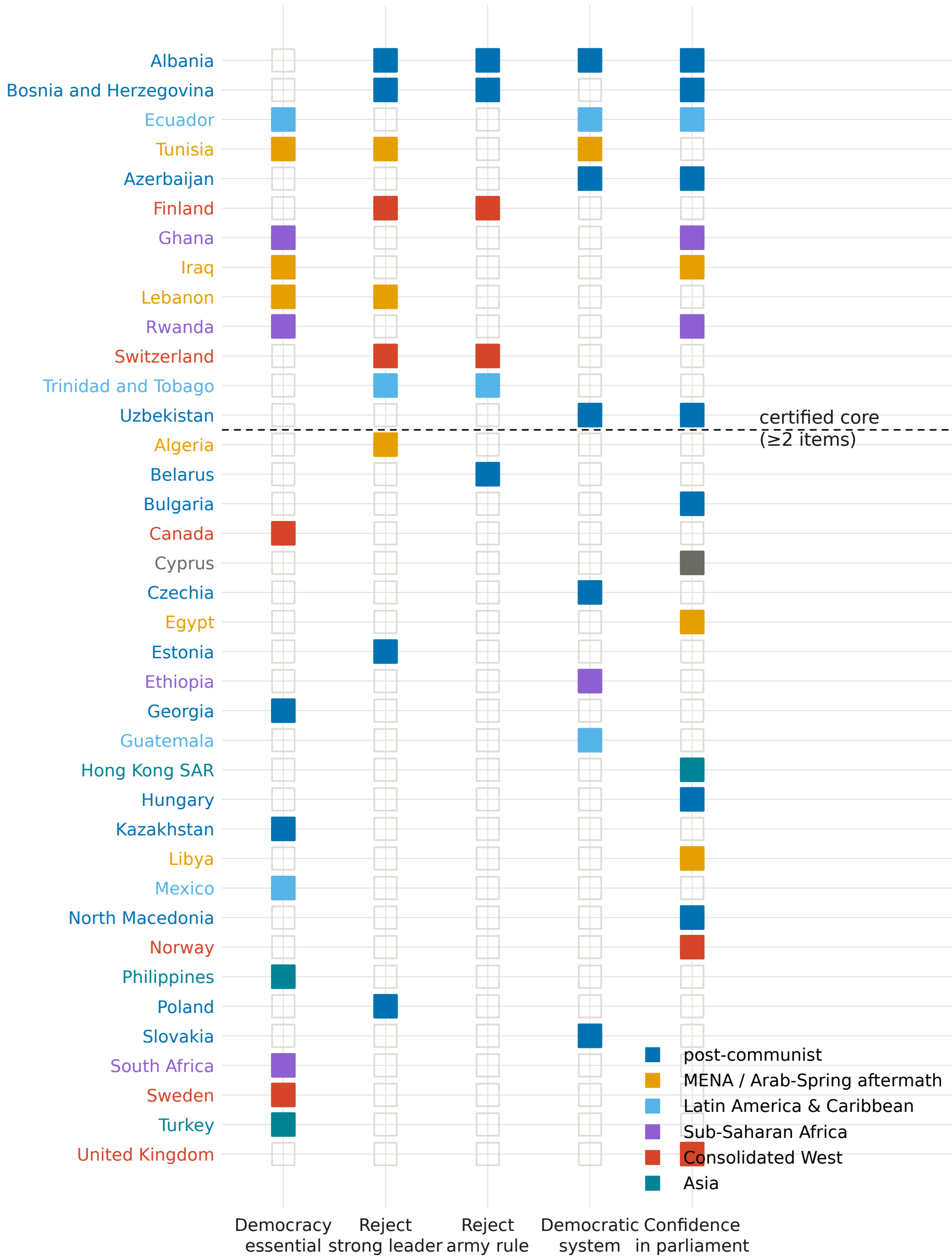

Figure S7: All thirty-eight countries certifying persistent distribution-wide decline on at least one battery item (WVS 1981–2022), colored by region; countries above the dashed line form the thirteen-country descriptive $\geq$ 2-item core shown, magnitude-aware, in the main text's Figure 4. Twelve also pass the arbitrary-cross-item-dependence partial-conjunction test, which certifies that at least two items decline without identifying which two.

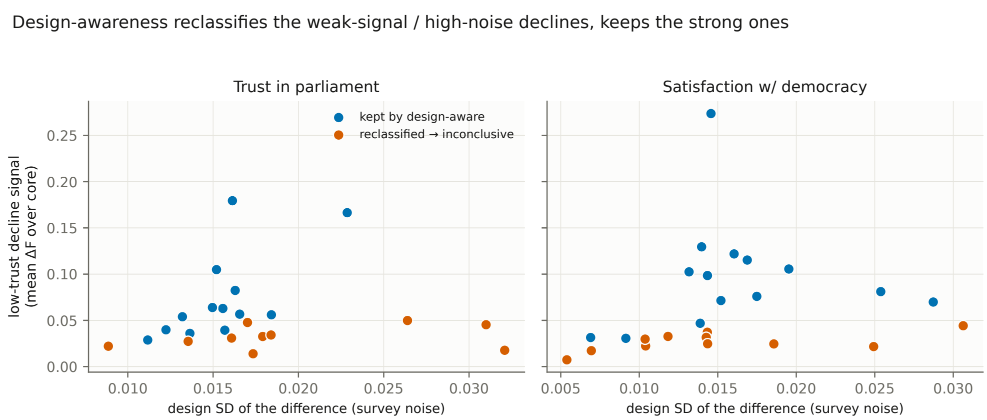

Figure S8: Relocated from the main text: countries certified by the marginal plug-in but inconclusive under the design-aware simultaneous band are the weak-signal, higher-noise cases; the apparent trend lies within the design uncertainty the survey warrants.